

1. Self Introduction

Question

Tell me about yourself.

Answer

Hello, my name is Vipin Kumar.

I have around 5 years of experience as a DevOps Engineer.

Currently, I am working on AWS cloud and Kubernetes environments.

My responsibilities include infrastructure setup, CI/CD pipeline creation, application deployment, monitoring, and production support.

I have hands-on experience with AWS, Docker, Kubernetes, Jenkins, Git, ArgoCD, Terraform, and Linux.

In my current project, we are deploying a three-tier application on AWS EKS.

I am currently looking for a new opportunity for career growth and technical improvement.

2. Current Project

Question

Explain your current project.

Answer

Currently, I am working on a three-tier application deployment project on AWS EKS.

The application contains frontend, backend, and database services.

Developers push code into Git repositories.

Jenkins pipelines are used for CI/CD automation. Kubernetes and ArgoCD

Docker images are created and pushed to the container registry.

are used for deployment automation.

My responsibilities include deployment, monitoring, troubleshooting, and infrastructure management.

3. CI/CD Pipeline

Question

Explain your CI/CD process.

Answer

Developers push code to Git repositories.

Jenkins automatically triggers the pipeline.

The application build process starts.

Docker images are created and pushed to the registry.

Kubernetes deployment files are updated.

ArgoCD detects changes from Git and deploys applications automatically to EKS clusters.

After deployment, we monitor the application health.

4. Docker

Question

What is Docker?

Answer

Docker is a containerization platform.

It is used to package applications and dependencies together.

Docker helps applications run consistently across different environments.

We use Docker to create lightweight and portable containers.

5. Docker Image vs Container**Question**

What is the difference between Docker image and container?

Answer

A Docker image is a template used to create containers.

It contains application code, libraries, and dependencies.

A container is a running instance of a Docker image.

Multiple containers can be created from a single image.

6. Git Branch**Question**

What is a branch in Git?

Answer

A branch in Git is an independent line of development.

It allows developers to work on features or bug fixes without affecting the main code.

After development and testing, changes are merged into the main branch.

7. Jenkins**Question**

What is Jenkins?

Answer

Jenkins is a CI/CD automation tool.

It is used to automate build, testing, and deployment processes.

Jenkins helps reduce manual work and improves deployment speed.

8. Kubernetes**Question**

What is Kubernetes?

Answer

Kubernetes is a container orchestration platform.

It is used to deploy, manage, and scale containerized applications.

Kubernetes provides features like auto-healing, scaling, and load balancing.

9. Pod

Question

What is a Pod?

Answer

A Pod is the smallest deployment unit in Kubernetes.

It contains one or more containers.

Containers inside a pod share network and storage resources.

10. Why Job Change

Question

Why are you looking for a job change?

Answer

I have learned many technologies in my current organization.

Now I am looking for better career growth and new technical challenges.

I also want to work on large-scale cloud and DevOps projects.

Day 2 — DevOps Interview Questions with Answers

Today focus on:

- AWS
- Kubernetes troubleshooting
- ArgoCD
- Terraform
- Linux
- English speaking flow

Speak slowly.

Pause after every sentence.

1. AWS EC2

Question

What is EC2 in AWS?

Answer

EC2 stands for Elastic Compute Cloud.

It is a virtual server service provided by AWS.

EC2 is used to run applications in the cloud.

We can configure CPU, memory, storage, and networking based on requirements.

2. IAM

Question

What is IAM?

Answer

IAM stands for Identity and Access Management.

It is used to manage users, roles, and permissions in AWS.

IAM helps provide secure access to AWS resources.

We can assign policies to users and roles.

3. VPC

Question

What is a VPC?

Answer

VPC stands for Virtual Private Cloud.

It provides an isolated network environment in AWS.

Inside a VPC, we can create subnets, route tables, and security groups.

It helps improve security and network management.

4. Security Group

Question

What is a Security Group?

Answer

A Security Group acts as a virtual firewall for AWS resources.

It controls inbound and outbound traffic.blockecific ports

We can allow or sp and IP addresses.

5. Load Balancer

Question

What is a Load Balancer?

Answer

A Load Balancer distributes incoming traffic across multiple servers or pods.

It helps improve application availability and performance.

In AWS, we commonly use Application Load Balancer and Network Load Balancer.

6. Kubernetes Deployment

Question

What is a Deployment in Kubernetes?

Answer

A Deployment is a Kubernetes resource used to manage applications.

It helps deploy and update pods automatically.
Deployments support scaling and rolling updates.

7. ReplicaSet

Question

What is a ReplicaSet?

Answer

A ReplicaSet ensures the required number of pod replicas are running.
If a pod fails, Kubernetes automatically creates a new pod.
It helps maintain application availability.

8. ConfigMap

Question

What is a ConfigMap?

Answer

A ConfigMap is used to store non-sensitive configuration data in Kubernetes.
It helps separate configuration from application code.
Applications can access ConfigMap values as environment variables or files.

9. Secret

Question

What is a Secret in Kubernetes?

Answer

A Secret is used to store sensitive information such as passwords, tokens, and API keys.
Kubernetes stores secrets in encoded format.
Applications can securely access secret values during runtime.

10. CrashLoopBackOff

Question

What is CrashLoopBackOff?

Answer

CrashLoopBackOff occurs when a container continuously fails and restarts.
Common reasons include application errors, wrong configurations, or missing dependencies.
We can troubleshoot using pod logs and describe commands.

11. ArgoCD

Question

What is ArgoCD?

Answer

ArgoCD is a GitOps continuous delivery tool for Kubernetes.

It automatically deploys applications from Git repositories.

ArgoCD continuously monitors Git and syncs changes to Kubernetes clusters.

12. GitOps

Question

What is GitOps?

Answer

GitOps is a deployment approach where Git acts as the single source of truth.

Infrastructure and deployment configurations are stored in Git repositories.

Tools like ArgoCD automatically apply changes from Git to Kubernetes clusters.

13. Terraform

Question

What is Terraform?

Answer

Terraform is an Infrastructure as Code tool.

It is used to automate infrastructure creation and management.

We can create AWS resources using configuration files.

14. Terraform Workflow

Question

Explain Terraform workflow.

Answer

First, we write Terraform configuration files.

Then we run terraform init to initialize the project.

After that, terraform plan is used to preview changes.

Finally, terraform apply creates the infrastructure.

15. Linux Command

Question

What is the use of grep command?

Answer

The grep command is used to search text patterns inside files.

It helps filter logs and find specific information quickly.

16. Linux Command

Question

What is top command?

Answer

The top command is used to monitor running processes and system resource usage. It shows CPU, memory, and process details in real time.

17. Scenario Question

Question

How do you troubleshoot a pod issue in Kubernetes?

Answer

First, I check pod status using `kubectl get pods`.
Then I check pod logs using `kubectl logs`.
After that, I use `kubectl describe pod` to identify events and errors.
I also verify resource usage, image issues, and configuration problems.
Finally, I fix the issue and restart the deployment if required.

18. Scenario Question

Question

What will you do if deployment fails in production?

Answer

First, I will identify the root cause from logs and monitoring tools.
I will check recent code or configuration changes.
If required, I will perform rollback to the previous stable version.
After fixing the issue, I will redeploy the application and monitor the environment.

English Speaking Practice

Speak Like This

“First, I check the logs.”
(pause)

“Then I identify the issue.”
(pause)

“After that, I fix the configuration.”
(pause)

Day 3 — Advanced DevOps Interview Questions with Answers

Today focus on:

- Kubernetes networking
- Ingress
- Helm
- Monitoring

- Production support
- Real-time scenarios

Speak slowly.

Pause after every sentence.

1. Service in Kubernetes

Question

What is a Service in Kubernetes?

Answer

A Service in Kubernetes is used to expose applications running inside pods.

It provides stable network communication between components.

Services help applications communicate even if pod IP addresses change.

2. Types of Kubernetes Services

Question

What are the different types of Services in Kubernetes?

Answer

Kubernetes provides multiple service types.

ClusterIP is used for internal communication.

NodePort exposes applications using node IP and port.

LoadBalancer exposes applications externally using cloud load balancers.

ExternalName maps services to external DNS names.

3. Ingress

Question

What is Ingress in Kubernetes?

Answer

Ingress is used to manage external access to applications in Kubernetes.

It provides routing based on domain names and paths.

Ingress helps manage HTTPS and load balancing efficiently.

4. Ingress Controller

Question

What is an Ingress Controller?

Answer

An Ingress Controller processes ingress rules and routes traffic to applications.

Common ingress controllers include NGINX and AWS Load Balancer Controller.

Without an ingress controller, ingress rules will not work.

5. Namespace

Question

What is a Namespace in Kubernetes?

Answer

A Namespace is used to logically separate Kubernetes resources.

It helps organize applications and environments.

We commonly create separate namespaces for development, staging, and production.

6. Helm

Question

What is Helm?

Answer

Helm is a package manager for Kubernetes.

It helps simplify application deployment and management.

Helm uses charts to manage Kubernetes resources efficiently.

7. Helm Chart

Question

What is a Helm Chart?

Answer

A Helm chart is a collection of Kubernetes YAML templates.

It contains deployment configurations, services, and values files.

Helm charts help standardize and automate deployments.

8. Rolling Update

Question

What is Rolling Update in Kubernetes?

Answer

Rolling update is a deployment strategy used to update applications gradually.

Old pods are replaced with new pods one by one.

This helps reduce downtime during deployments.

9. Monitoring

Question

What monitoring tools have you used?

Answer

I have worked with CloudWatch, Prometheus, and Grafana.

Prometheus is used to collect metrics.

Grafana is used for dashboards and visualization.

CloudWatch is used for AWS monitoring and alerts.

10. Prometheus

Question

What is Prometheus?

Answer

Prometheus is an open-source monitoring tool.

It collects metrics from applications and infrastructure.

Prometheus uses pull-based monitoring and supports alerting.

11. Grafana

Question

What is Grafana?

Answer

Grafana is a visualization tool used for monitoring dashboards.

It connects with Prometheus and other data sources.

Grafana helps visualize CPU, memory, and application metrics.

12. Scenario Question

Question

How do you monitor application health in production?

Answer

We use monitoring tools like Prometheus and Grafana.

We monitor CPU, memory, pod health, and application response times.

Alerts are configured for critical issues.

We also check logs and dashboards regularly.

13. Production Support

Question

Have you worked on production support?

Answer

Yes, I have experience in production support.

I monitor deployments, troubleshoot issues, and handle alerts.

I also perform rollback and coordinate with development teams during incidents.

14. Scenario Question

Question

What will you do if pod CPU usage is very high?

Answer

First, I will check pod metrics and logs.

Then I will identify whether the issue is application-related or traffic-related.

I can increase resource limits or scale the deployment if required.

I will also verify if there are memory leaks or abnormal processes.

15. Scenario Question

Question

How do you perform rollback in Kubernetes?

Answer

Kubernetes supports deployment rollback using rollout history.

We can rollback to the previous stable version using `kubectl rollout undo` command.

Rollback helps restore application stability quickly.

16. EKS

Question

Why are you using EKS?

Answer

EKS is a managed Kubernetes service provided by AWS.

It reduces operational overhead for Kubernetes cluster management.

AWS manages the control plane, scalability, and high availability.

17. High Availability

Question

What is High Availability?

Answer

High availability means applications remain accessible even during failures.

We achieve this using multiple replicas, load balancing, and multi-zone deployment.

High availability helps reduce downtime.

18. DevOps Culture

Question

What is DevOps?

Answer

DevOps is a culture that improves collaboration between development and operations teams.

It focuses on automation, continuous integration, continuous deployment, and faster delivery.

DevOps helps improve deployment speed and reliability.

19. Troubleshooting Flow

Question

How do you troubleshoot application downtime?

Answer

First, I check monitoring alerts and logs.

Then I verify pod and node status.

I identify recent deployment or configuration changes.

After finding the root cause, I fix the issue and monitor application stability.

English Speaking Practice

Practice Slowly

“First, I check the logs.”

(pause)

“Then I identify the root cause.”

(pause)

“After that, I fix the issue.”

(pause)

Day 4 — Real-Time DevOps Scenario Questions with Answers

Today focus on:

- real production issues
- troubleshooting
- Kubernetes debugging
- Jenkins failures
- AWS scenarios
- interview confidence

These questions are very common for 5 years experienced DevOps engineers.

Speak slowly.

Pause after every sentence.

1. Scenario — Pod Not Starting

Question

A pod is stuck in Pending state. How will you troubleshoot?

Answer

First, I will check the pod details using `kubectl describe pod` command.

Then I will verify node resources like CPU and memory.

I will also check whether the required node labels and taints are configured properly.

After that, I will verify PVC and storage-related issues.

Finally, I will fix the issue and monitor pod status.

2. Scenario — CrashLoopBackOff

Question

How do you troubleshoot CrashLoopBackOff?

Answer

First, I check pod logs using `kubectl logs` command.

Then I verify environment variables, secrets, and config maps.

I also check application startup errors and database connectivity.

After identifying the root cause, I fix the configuration or application issue.

Finally, I restart the deployment and monitor pod health.

3. Scenario — ImagePullBackOff

Question

What is ImagePullBackOff?

Answer

ImagePullBackOff occurs when Kubernetes cannot pull the container image.

Common reasons include incorrect image name, authentication failure, or missing image.

I verify image repository access and image tags.

Then I update the deployment configuration and restart the pod.

4. Scenario — Jenkins Pipeline Failure

Question

What will you do if Jenkins pipeline fails?

Answer

First, I check Jenkins console logs to identify the failure stage.

Then I verify build errors, connectivity issues, or permission problems.

I also check Git access, Docker build process, and deployment configuration.

After fixing the issue, I rerun the pipeline and monitor deployment status.

5. Scenario — Application Not Accessible

Question

Application is deployed but not accessible. How will you troubleshoot?

Answer

First, I verify pod and service status.

Then I check ingress and load balancer configuration.

I also verify security groups, DNS, and network policies.

After that, I check application logs and port mappings.

Finally, I test connectivity and confirm application accessibility.

6. Scenario — High Memory Usage

Question

What will you do if memory usage is very high in Kubernetes?

Answer

First, I check pod metrics and resource utilization.

Then I identify which application is consuming high memory.

I verify memory limits and requests configuration.

If required, I increase resources or scale the deployment.

I also coordinate with developers if there is a memory leak issue.

7. Scenario — Node Not Ready

Question

How do you troubleshoot a node in NotReady state?

Answer

First, I check node status using `kubectl get nodes`.

Then I verify kubelet service and node connectivity.

I also check CPU, memory, and disk utilization.

After that, I review node logs and networking issues.

Finally, I restart services or recover the node if required.

8. Scenario — Deployment Rollback

Question

How do you rollback a failed deployment?

Answer

Kubernetes supports rollback using rollout history.

First, I identify the failed deployment version.

Then I rollback to the previous stable version using `kubectl rollout undo` command.

After rollback, I verify application stability and monitor logs.

9. Scenario — Database Connectivity Issue

Question

Application cannot connect to database. What will you check?

Answer

First, I verify database pod or server status.

Then I check connection strings and secrets configuration.

I also verify network connectivity and firewall rules.

After that, I review application logs for database-related errors.

Finally, I fix the configuration and test connectivity.

10. Scenario — Disk Space Full

Question

Linux server disk space is full. What will you do?

Answer

First, I check disk usage using `df -h` command.

Then I identify large files and logs using `du` command.

I clean unnecessary logs and temporary files.

If required, I increase storage capacity.

Finally, I monitor server health after cleanup.

11. Scenario — SSL Certificate Expired

Question

What will you do if SSL certificate expires?

Answer

First, I verify the certificate expiration details.

Then I renew or replace the SSL certificate.

After that, I update ingress or load balancer configuration.

Finally, I restart services if required and test HTTPS access.

12. Scenario — ArgoCD Sync Failed

Question

What will you do if ArgoCD sync fails?

Answer

First, I check ArgoCD application events and logs.

Then I verify Git repository access and Kubernetes permissions.

I also check YAML syntax and deployment errors.

After fixing the issue, I trigger sync again and verify deployment status.

13. Scenario — High Traffic

Question

How do you handle sudden high traffic?

Answer

We use auto-scaling and load balancing to handle high traffic.

Kubernetes HPA automatically increases pod replicas based on CPU or memory usage.

AWS load balancers distribute traffic efficiently across pods.

We continuously monitor application performance during traffic spikes.

14. HPA

Question

What is HPA in Kubernetes?

Answer

HPA stands for Horizontal Pod Autoscaler.

It automatically scales pod replicas based on resource usage.

HPA helps improve application availability during high traffic.

15. Logging

Question

How do you manage logs in Kubernetes?

Answer

We collect logs using centralized logging solutions.

Logs help identify application and infrastructure issues.

We monitor logs regularly for troubleshooting and debugging.

16. Production Deployment

Question

How do you ensure safe production deployment?

Answer

We use CI/CD pipelines with automated testing.

Rolling updates help reduce downtime during deployment.

We also monitor application health after deployment.

Rollback strategy is prepared in case of failures.

Speaking Practice

Practice This Style

“First, I check the logs.”

(pause)

“Then I identify the issue.”

(pause)

“After that, I fix the configuration.”

(pause)

Day 5 — Advanced Kubernetes + AWS + DevOps Interview Questions

Today focus on:

- advanced Kubernetes
- networking
- security
- CI/CD

- AWS architecture
- real production experience

These questions are commonly asked for 5 years experience.

Speak slowly.

Pause after every sentence.

1. StatefulSet

Question

What is StatefulSet in Kubernetes?

Answer

StatefulSet is used for stateful applications in Kubernetes.

It provides stable network identity and persistent storage.

StatefulSet is commonly used for databases like MySQL and MongoDB.

Each pod gets a unique name and persistent volume.

2. DaemonSet

Question

What is DaemonSet?

Answer

DemonSet ensures one pod runs on every node.

It is commonly used for monitoring and logging agents.

Examples include Fluentd and node exporters.

3. Persistent Volume

Question

What is Persistent Volume in Kubernetes?

Answer

Persistent Volume is a storage resource in Kubernetes.

It provides persistent storage for applications.

Data remains available even if pods restart.

4. Persistent Volume Claim

Question

What is PVC?

Answer

PVC stands for Persistent Volume Claim.

It is used by applications to request storage resources.

Kubernetes automatically binds PVC with suitable persistent volumes.

5. Resource Limits

Question

Why do we use resource limits in Kubernetes?

Answer

Resource limits help control CPU and memory usage for containers.

They prevent one application from consuming excessive resources.

Resource requests help Kubernetes schedule pods efficiently.

6. Taints and Tolerations

Question

What are taints and tolerations?

Answer

Taints are applied on nodes to restrict pod scheduling.

Tolerations allow pods to run on tainted nodes.

They help control workload placement in Kubernetes clusters.

7. Node Selector

Question

What is Node Selector?

Answer

Node Selector is used to schedule pods on specific nodes.

It works based on node labels.

This helps run workloads on dedicated nodes.

8. Blue-Green Deployment

Question

What is Blue-Green Deployment?

Answer

Blue-Green deployment is a release strategy used to reduce downtime.

Two environments are maintained simultaneously.

One environment runs the current version and another runs the new version.

Traffic is switched after successful testing.

9. Canary Deployment

Question

What is Canary Deployment?

Answer

Canary deployment releases the new version to a small percentage of users first.

It helps test application stability before full deployment.
This reduces deployment risk in production environments.

10. Infrastructure as Code

Question

What is Infrastructure as Code?

Answer

Infrastructure as Code is the process of managing infrastructure using code files.
Tools like Terraform automate infrastructure provisioning.
It improves consistency, automation, and version control.

11. Terraform State File

Question

What is Terraform state file?

Answer

Terraform state file stores infrastructure information.
It helps Terraform track resource changes.
Remote state storage is commonly used for team collaboration.

12. Remote Backend

Question

Why do we use remote backend in Terraform?

Answer

Remote backend stores Terraform state centrally.
It improves team collaboration and state management.
AWS S3 and DynamoDB are commonly used for remote backend configuration.

13. Route53

Question

What is Route53?

Answer

Route53 is a DNS service provided by AWS.
It is used for domain registration and traffic routing.
Route53 supports health checks and failover routing.

14. Auto Scaling

Question

What is Auto Scaling in AWS?

Answer

Auto Scaling automatically increases or decreases EC2 instances based on traffic.

It helps improve availability and optimize costs.

Auto Scaling works with CloudWatch alarms and scaling policies.

15. S3**Question**

What is S3?

Answer

S3 stands for Simple Storage Service.

It is an object storage service provided by AWS.

S3 is used for backups, static websites, and artifact storage.

16. CloudWatch**Question**

What is CloudWatch?

Answer

CloudWatch is a monitoring service in AWS.

It collects logs, metrics, and alerts from AWS resources.

CloudWatch helps monitor infrastructure and application health.

17. Security Best Practices**Question**

How do you improve security in Kubernetes?

Answer

We use RBAC for access control.

Secrets are used for sensitive information.

Network policies help restrict traffic.

We also scan container images for vulnerabilities.

Regular updates and monitoring improve cluster security.

18. Container Security**Question**

How do you scan Docker images?

Answer

We use tools like Trivy for container image scanning.

It helps identify vulnerabilities in Docker images.

Scanning is integrated into CI/CD pipelines before deployment.

19. Scenario Question

Question

How do you reduce deployment downtime?

Answer

We use rolling updates and load balancing.

Health checks ensure only healthy pods receive traffic.

Canary and blue-green deployments help reduce production risk.

Rollback strategy is also prepared before deployment.

20. Scenario Question

Question

How do you secure CI/CD pipelines?

Answer

We use role-based access control and secure credentials management.

Secrets are stored securely and not hardcoded in pipelines.

We also perform image scanning and access auditing.

Pipeline permissions are restricted based on user roles.

21. Scenario Question

Question

What challenges have you faced in production?

Answer

I have handled deployment failures, pod crashes, and high resource usage issues.

I troubleshoot logs, monitor alerts, and coordinate with development teams.

I also perform rollback and infrastructure optimization when required.

22. Leadership Question

Question

Have you worked with teams?

Answer

Yes, I regularly work with development, QA, and operations teams.

I coordinate deployment activities and production issue resolution.

Collaboration is important for successful DevOps implementation.

English Speaking Practice

Practice This Slowly

“First, I identify the issue.”

(pause)

“Then I check the logs and monitoring dashboards.”

(pause)

“After that, I fix the configuration.”

(pause)

Day 6 — Jenkins + Linux + Networking + Scenario Interview Questions

Today focus on:

- Jenkins pipelines
- Linux troubleshooting
- Networking basics
- DNS
- Load balancer
- Production debugging

These are very common in DevOps interviews.

Speak slowly.

Pause after every sentence.

1. Jenkins Pipeline

Question

What is a Jenkins pipeline?

Answer

Jenkins pipeline is a CI/CD automation workflow.

It defines build, test, and deployment stages.

Pipelines help automate software delivery processes.

2. Declarative vs Scripted Pipeline

Question

What is the difference between Declarative and Scripted pipeline?

Answer

Declarative pipeline uses predefined syntax and is easier to manage.

Scripted pipeline provides more flexibility using Groovy scripting.

Declarative pipelines are commonly used for standard CI/CD workflows.

3. Jenkinsfile

Question

What is Jenkinsfile?

Answer

Jenkinsfile is a text file that defines Jenkins pipeline stages.

It is stored in the Git repository.

Jenkins reads the Jenkinsfile and executes pipeline steps automatically.

4. CI/CD Benefits

Question

What are the benefits of CI/CD?

Answer

CI/CD improves deployment speed and automation.

It reduces manual errors and improves software quality.

CI/CD also supports faster release cycles and continuous delivery.

5. Linux — Top Command

Question

What is the use of top command?

Answer

The top command displays running processes and system resource usage.

It helps monitor CPU and memory utilization in real time.

6. Linux — ps Command

Question

What is ps command?

Answer

The ps command displays running processes in Linux.

It helps identify process IDs and application status.

7. Linux — grep Command

Question

What is grep command?

Answer

grep is used to search specific patterns inside files.

It is commonly used for log analysis and troubleshooting.

8. Linux — chmod

Question

What is chmod command?

Answer

chmod is used to change file permissions in Linux.

It controls read, write, and execute access for users and groups.

9. Linux — systemctl

Question

What is systemctl?

Answer

systemctl is used to manage Linux services.

It can start, stop, restart, and check service status.

10. DNS**Question**

What is DNS?

Answer

DNS stands for Domain Name System.

It converts domain names into IP addresses.

DNS helps users access applications using domain names instead of IP addresses.

11. Load Balancer**Question**

Why do we use Load Balancer?

Answer

Load balancer distributes traffic across multiple servers or pods.

It improves application availability and performance.

Load balancing helps prevent server overload.

12. HTTP vs HTTPS**Question**

What is the difference between HTTP and HTTPS?

Answer

HTTP is an unsecured communication protocol.

HTTPS provides encrypted communication using SSL or TLS certificates.

HTTPS improves security for web applications.

13. TCP vs UDP**Question**

What is the difference between TCP and UDP?

Answer

TCP is a connection-oriented protocol.

It provides reliable communication and error checking.

UDP is connectionless and faster than TCP.

UDP is commonly used for streaming and gaming applications.

14. Port 80 and 443

Question

What are ports 80 and 443?

Answer

Port 80 is used for HTTP traffic.

Port 443 is used for HTTPS secure traffic.

15. Scenario Question

Question

Website is slow. How will you troubleshoot?

Answer

First, I check CPU, memory, and disk usage.

Then I verify application logs and monitoring dashboards.

I also check database performance and network latency.

After identifying the issue, I optimize resources or scale the application.

16. Scenario Question

Question

How do you troubleshoot high CPU usage?

Answer

First, I identify high CPU-consuming processes using top command.

Then I verify application logs and traffic patterns.

I restart unhealthy services if required.

I also scale the application if traffic is high.

17. Scenario Question

Question

Jenkins is not accessible. What will you check?

Answer

First, I verify whether the Jenkins service is running.

Then I check server CPU, memory, and disk usage.

I also verify firewall rules and network connectivity.

After that, I review Jenkins logs for errors.

18. Scenario Question

Question

How do you troubleshoot network issues in Kubernetes?

Answer

First, I verify pod and service connectivity.

Then I check DNS resolution and ingress configuration.

I also verify network policies and security groups.

After that, I test connectivity using curl or ping commands.

19. Scenario Question**Question**

How do you handle production deployment pressure?

Answer

I stay calm and follow standard deployment procedures.

I verify backups and rollback plans before deployment.

During incidents, I coordinate with teams and troubleshoot issues quickly.

Monitoring and communication are very important during production deployments.

20. Team Collaboration**Question**

How do you work with developers?

Answer

I work closely with developers during deployment and troubleshooting activities.

I help resolve infrastructure and CI/CD-related issues.

Good communication and collaboration improve delivery speed.

21. Monitoring Alerts**Question**

What will you do if you receive a production alert at night?

Answer

First, I check monitoring dashboards and logs.

Then I identify whether the issue is critical or temporary.

I troubleshoot the issue and restore application stability.

If required, I escalate the issue to the appropriate team.

22. Logging**Question**

Why are logs important?

Answer

Logs help identify application and infrastructure issues.

They provide error details and troubleshooting information.

Logs are very important for debugging production problems.

Day 7 — Advanced AWS + Kubernetes + DevOps Scenario Questions

Today focus on:

- AWS architecture
- Kubernetes production issues
- Disaster recovery
- Security
- Scaling
- Real-time DevOps scenarios

These questions are frequently asked for experienced DevOps engineers.

Speak slowly.

Pause after every sentence.

1. AWS Architecture

Question

Explain your AWS architecture.

Answer

Our application is deployed on AWS cloud.

We are using a VPC with public and private subnets.

Load balancers are placed in public subnets.

Application workloads run inside EKS clusters in private subnets.

Route53 is used for DNS management.

CloudWatch is used for monitoring and alerts.

We also use IAM roles for secure access management.

2. Public vs Private Subnet

Question

What is the difference between public and private subnet?

Answer

Public subnets have internet access through an internet gateway.

Resources like load balancers are usually placed in public subnets.

Private subnets do not have direct internet access.

Application and database servers are commonly deployed in private subnets for security.

3. NAT Gateway

Question

What is a NAT Gateway?

Answer

NAT Gateway allows private subnet resources to access the internet securely.

It is commonly used for downloading packages and updates without exposing servers publicly.

4. IAM Role

Question

What is an IAM Role?

Answer

IAM role provides temporary permissions to AWS services or users.

Roles improve security by avoiding hardcoded credentials.

EC2 and EKS commonly use IAM roles for AWS access.

5. EKS Worker Node

Question

What are worker nodes in EKS?

Answer

Worker nodes are EC2 instances that run Kubernetes pods.

They join the EKS cluster and execute application workloads.

Kubernetes schedules pods on worker nodes based on resource availability.

6. Cluster Autoscaler

Question

What is Cluster Autoscaler?

Answer

Cluster Autoscaler automatically increases or decreases worker nodes based on workload demand.

It helps optimize infrastructure cost and application availability.

7. Horizontal Pod Autoscaler

Question

How does HPA work?

Answer

HPA monitors CPU or memory usage of pods.

When resource usage increases, Kubernetes automatically increases pod replicas.

When traffic decreases, replicas are reduced automatically.

8. Kubernetes Secret

Question

How do you manage secrets in Kubernetes?

Answer

Kubernetes secrets are used to store sensitive data securely.

Secrets include passwords, API keys, and tokens.

Applications access secrets as environment variables or mounted files.

9. RBAC

Question

What is RBAC in Kubernetes?

Answer

RBAC stands for Role-Based Access Control.

It controls user and service permissions inside Kubernetes clusters.

RBAC improves security by restricting unauthorized access.

10. Production Issue

Question

A node suddenly goes down. What happens?

Answer

Kubernetes automatically reschedules pods to healthy nodes.

ReplicaSets ensure the required number of pods are running.

Monitoring alerts help identify node failures quickly.

11. Disaster Recovery

Question

What is Disaster Recovery?

Answer

Disaster recovery is the process of restoring systems after failures.

Backups, replication, and multi-region setups improve recovery capability.

Disaster recovery helps reduce downtime and data loss.

12. Backup Strategy

Question

How do you manage backups?

Answer

We take regular backups of databases and important configurations.

S3 is commonly used for backup storage.

Backup validation and recovery testing are also important.

13. Kubernetes Upgrade

Question

How do you upgrade Kubernetes clusters?

Answer

First, we review compatibility and backup configurations.
Then we upgrade the control plane and worker nodes gradually.
We monitor application stability during the upgrade process.
Rollback planning is also prepared before upgrades.

14. Container Restart**Question**

Why do containers restart frequently?

Answer

Common reasons include application crashes, insufficient resources, or failed health checks.
I troubleshoot logs and resource usage to identify the root cause.

15. Readiness vs Liveness Probe**Question**

What is the difference between readiness and liveness probe?

Answer

Readiness probe checks whether the application is ready to receive traffic.
Liveness probe checks whether the application is healthy or stuck.
Kubernetes restarts containers if liveness probe fails.

16. Scenario Question**Question**

How do you handle zero downtime deployment?

Answer

We use rolling updates and load balancing.
Health checks ensure traffic goes only to healthy pods.
Blue-green or canary deployment strategies are also used for safer releases.

17. Scenario Question**Question**

Database storage is full. What will you do?

Answer

First, I verify storage utilization and logs.
Then I clean unnecessary files and backups if possible.
If required, I increase storage capacity.
I also coordinate with database teams for optimization.

18. Scenario Question

Question

How do you troubleshoot DNS issues in Kubernetes?

Answer

First, I verify CoreDNS pod status.

Then I check service names and namespace configuration.

I also test DNS resolution from inside the pod.

After identifying the issue, I fix DNS or networking configuration.

19. Security Question

Question

How do you secure Docker containers?

Answer

We use minimal base images and vulnerability scanning.

Containers run with limited permissions.

Secrets are managed securely.

Regular image updates improve security.

20. DevOps Best Practice

Question

What are DevOps best practices?

Answer

Automation is very important in DevOps.

Infrastructure should be managed as code.

CI/CD pipelines improve delivery speed.

Monitoring and security should be integrated into deployment workflows.

21. Leadership Question

Question

Have you handled critical incidents?

Answer

Yes, I have handled deployment failures and production outages.

I troubleshoot issues, coordinate with teams, and restore services quickly.

Calm communication and monitoring are important during incidents.

Day 8 — HR + Behavioral + Manager Round Questions with Answers

Today focus on:

- HR round
- confidence

- communication
- leadership
- team handling
- difficult situations

These questions are very important for experienced candidates.

Speak slowly.

Pause after every sentence.

1. Tell Me About Yourself

Question

Tell me about yourself.

Answer

Hello, my name is Vipin Kumar.

I have around 5 years of experience as a DevOps Engineer.

I have hands-on experience with AWS, Kubernetes, Docker, Jenkins, Terraform, Git, and ArgoCD.

Currently, I am working on deployment automation and Kubernetes infrastructure management on AWS EKS.

My responsibilities include CI/CD pipeline creation, monitoring, troubleshooting, and production support.

I am looking for a new opportunity to improve my technical skills and career growth.

2. Why Do You Want to Change Your Job?

Question

Why are you looking for a job change?

Answer

I have learned many technologies in my current organization.

Now I am looking for better career growth and new challenges.

I want to work on large-scale cloud and DevOps projects where I can improve my technical and professional skills.

3. Why Should We Hire You?

Question

Why should we hire you?

Answer

I have strong hands-on experience in AWS and Kubernetes environments.

I have worked on CI/CD automation, troubleshooting, and production support.

I can quickly identify and resolve deployment issues.

I am also a quick learner and work well with teams.

4. What Are Your Strengths?

Question

What are your strengths?

Answer

My strengths are troubleshooting, learning new technologies, and handling production issues calmly.

I also have good experience in automation and Kubernetes deployments.

I work well under pressure and support team collaboration.

5. What Is Your Weakness?**Question**

What is your weakness?

Answer

Earlier, I was not very confident in communication during interviews.

Now I am continuously improving my speaking skills and practicing regularly.

I always focus on self-improvement and learning.

6. Team Conflict**Question**

Have you faced conflict with team members?

Answer

Yes, sometimes technical discussions create different opinions.

I try to understand everyone's perspective calmly.

We discuss the issue and choose the best solution based on project requirements.

Good communication helps resolve conflicts professionally.

7. Pressure Handling**Question**

How do you handle pressure?

Answer

I stay calm and focus on identifying the root cause.

I prioritize critical tasks and communicate clearly with teams.

During production issues, structured troubleshooting helps reduce stress.

8. Leadership**Question**

Have you handled responsibilities independently?

Answer

Yes, I have handled deployments, troubleshooting, and production support independently.

I also coordinate with development and QA teams during releases.

I take ownership of issues and ensure timely resolution.

9. Biggest Challenge

Question

What was the biggest challenge you faced?

Answer

One major challenge was handling a production deployment failure.

The application pods were crashing after deployment.

I checked logs and identified a configuration issue related to database connectivity.

After fixing the configuration and performing rollback, the application became stable.

10. Production Outage

Question

Have you handled production outages?

Answer

Yes, I have handled production incidents and deployment failures.

I monitor alerts, troubleshoot issues, and coordinate with teams for resolution.

Fast troubleshooting and communication are very important during outages.

11. Work Under Deadline

Question

Can you work under tight deadlines?

Answer

Yes, I can work under deadlines.

In DevOps environments, production deployments and incidents often require quick action.

I stay focused and prioritize tasks effectively.

12. Career Goals

Question

What are your career goals?

Answer

My goal is to become a highly skilled cloud and DevOps professional.

I want to improve my expertise in Kubernetes, cloud architecture, and automation.

I also want to contribute to large-scale production environments.

13. Why DevOps?

Question

Why did you choose DevOps?

Answer

I am interested in automation, cloud technologies, and infrastructure management.

DevOps combines development, operations, and automation together.

I enjoy solving deployment and infrastructure challenges.

14. Learning New Technology

Question

How do you learn new technologies?

Answer

I learn through documentation, hands-on practice, and real project experience.

I also watch technical videos and practice in lab environments.

Continuous learning is very important in DevOps.

15. Handling Mistakes

Question

Have you ever made a mistake in production?

Answer

Yes, mistakes can happen in production environments.

One time, a deployment configuration issue caused application downtime.

I identified the issue quickly, performed rollback, and restored services.

After that, we improved validation and deployment checks.

16. Communication

Question

How do you communicate with teams during incidents?

Answer

I provide regular updates and explain the issue clearly.

I coordinate with developers, QA, and management teams during incidents.

Clear communication helps reduce confusion during critical situations.

17. Salary Discussion

Question

What are your salary expectations?

Answer

I am looking for a salary based on my experience, technical skills, and market standards.

I am open to discussing the compensation based on the role and responsibilities.

18. Relocation

Question

Are you open to relocation?

Answer

Yes, I am open to relocation based on project and company requirements.

19. Notice Period**Question**

What is your notice period?

Answer

My current notice period is _____.

However, I will try my best to join as early as possible.

20. Final HR Question**Question**

Do you have any questions for us?

Answer

Yes.

I would like to know more about the team structure and current projects.

I also want to understand the technologies and deployment environments used in this role.

Day 9 — Kubernetes Deep-Dive Interview Questions with Answers

Today focus on:

- advanced Kubernetes
- networking
- scheduling
- security
- cluster internals
- production troubleshooting

These questions are commonly asked for experienced DevOps engineers.

Speak slowly.

Pause after every sentence.

1. Kubernetes Architecture**Question**

Explain Kubernetes architecture.

Answer

Kubernetes architecture contains control plane components and worker nodes.

The control plane manages the cluster.

Components include API Server, Scheduler, Controller Manager, and etcd.

Worker nodes run application pods using kubelet and container runtime.

2. API Server

Question

What is the API Server?

Answer

API Server is the main entry point of the Kubernetes cluster.

It receives and processes all cluster requests.

kubectl commands communicate with the API Server.

3. etcd

Question

What is etcd?

Answer

etcd is a distributed key-value database used by Kubernetes.

It stores cluster configuration and state information.

Kubernetes uses etcd for cluster consistency and recovery.

4. Scheduler

Question

What is Kubernetes Scheduler?

Answer

Scheduler decides on which worker node a pod should run.

It checks resource availability and scheduling policies before assigning pods.

5. Controller Manager

Question

What is Controller Manager?

Answer

Controller Manager manages Kubernetes controllers.

Controllers continuously monitor cluster state and maintain the desired state.

Examples include ReplicaSet and Node controllers.

6. Kubelet

Question

What is Kubelet?

Answer

Kubelet is an agent running on worker nodes.

It communicates with the API Server and manages pod execution.

Kubelet ensures containers are running properly.

7. Container Runtime

Question

What is Container Runtime?

Answer

Container runtime is responsible for running containers on worker nodes.
Common runtimes include containerd and CRI-O.

8. Node Affinity

Question

What is Node Affinity?

Answer

Node Affinity controls pod scheduling based on node labels.
It provides advanced scheduling rules compared to node selectors.

9. Network Policy

Question

What is Network Policy?

Answer

Network Policy controls traffic flow between pods.
It improves cluster security by restricting unauthorized communication.

10. Init Container

Question

What is an Init Container?

Answer

Init containers run before the main application container starts.
They are commonly used for initialization tasks and dependency checks.

11. Sidecar Container

Question

What is a Sidecar Container?

Answer

Sidecar containers run alongside the main application container.
They provide additional functionality such as logging and monitoring.

12. Stateful Application

Question

Why do databases use StatefulSet instead of Deployment?

Answer

Databases require persistent storage and stable network identity.

StatefulSet provides fixed pod names and persistent volumes.

Deployments are mainly used for stateless applications.

13. Pod Communication**Question**

How do pods communicate in Kubernetes?

Answer

Pods communicate using Kubernetes networking.

Each pod receives a unique IP address.

Services provide stable communication between pods.

14. DNS in Kubernetes**Question**

How does DNS work in Kubernetes?

Answer

Kubernetes uses CoreDNS for service discovery.

Services are accessible using service names inside the cluster.

DNS automatically resolves service names into cluster IP addresses.

15. Service Discovery**Question**

What is Service Discovery?

Answer

Service discovery helps applications locate other services dynamically.

Kubernetes DNS automatically provides service discovery inside the cluster.

16. Resource Requests vs Limits**Question**

What is the difference between resource requests and limits?

Answer

Resource requests define the minimum resources required by a container.

Limits define the maximum resources a container can use.

Kubernetes scheduler uses requests for pod placement.

17. OOMKilled**Question**

What is OOMKilled?

Answer

OOMKilled occurs when a container exceeds its memory limit.

Kubernetes terminates the container to protect node stability.

Increasing memory limits or optimizing the application can resolve the issue.

18. Multi-Container Pod

Question

Why do we use multi-container pods?

Answer

Multi-container pods allow closely related containers to work together.

Containers inside the same pod share network and storage resources.

19. Kubernetes Logging

Question

How do you check Kubernetes logs?

Answer

We use `kubectl logs` command to check pod logs.

Centralized logging solutions are also used for production environments.

20. Kubernetes Events

Question

How do you check Kubernetes events?

Answer

We use `kubectl get events` command to view cluster events.

Events help identify scheduling and deployment issues.

21. Scenario Question

Question

A pod is running but application is not working. What will you check?

Answer

First, I verify pod logs and application health checks.

Then I check service configuration and port mappings.

I also verify database connectivity and environment variables.

Finally, I test network connectivity and application endpoints.

22. Scenario Question

Question

Pods are not scheduling. How will you troubleshoot?

Answer

First, I check node resource availability.

Then I verify taints, tolerations, and affinity rules.

I also check events using `kubectl describe pod` command.

After identifying the issue, I update resource or scheduling configuration.

23. Scenario Question

Question

How do you troubleshoot Kubernetes networking issues?

Answer

First, I verify pod and service connectivity.

Then I check DNS resolution and ingress configuration.

I also verify network policies and firewall rules.

Finally, I test communication between pods using `curl` or `ping` commands.

24. Scenario Question

Question

How do you secure Kubernetes clusters?

Answer

We use RBAC for access control.

Secrets are used for sensitive information.

Network policies restrict unnecessary communication.

Image scanning and regular updates improve cluster security.

Day 10 — AWS Deep-Dive Interview Questions with Answers

Today focus on:

- AWS networking
- security
- scaling
- storage
- monitoring
- production troubleshooting

These questions are very common for experienced DevOps engineers.

Speak slowly.

Pause after every sentence.

1. AWS Regions and Availability Zones

Question

What are Regions and Availability Zones in AWS?

Answer

AWS Regions are geographical locations containing AWS data centers.

Each region contains multiple Availability Zones.

Availability Zones are isolated data centers within a region.

They help improve high availability and fault tolerance.

2. Internet Gateway

Question

What is an Internet Gateway?

Answer

Internet Gateway allows communication between AWS resources and the internet.

It is attached to a VPC.

Public subnets use Internet Gateway for internet access.

3. Route Table

Question

What is a Route Table?

Answer

Route Table controls network traffic inside a VPC.

It contains routing rules for subnet communication.

Routes define where network traffic should go.

4. Security Group vs NACL

Question

What is the difference between Security Group and NACL?

Answer

Security Group works at the instance level.

It acts as a stateful firewall.

NACL works at the subnet level.

It acts as a stateless firewall.

5. EC2 Instance Types

Question

What are EC2 instance types?

Answer

EC2 instance types provide different CPU, memory, and storage configurations.

Different instance families are optimized for different workloads.

Examples include general purpose, compute optimized, and memory optimized instances.

6. AMI

Question

What is an AMI?

Answer

AMI stands for Amazon Machine Image.

It is a template used to launch EC2 instances.

AMIs contain operating systems and application configurations.

7. EBS

Question

What is EBS?

Answer

EBS stands for Elastic Block Store.

It provides persistent block storage for EC2 instances.

EBS volumes remain available even after instance restart.

8. S3 Storage Classes

Question

What are S3 storage classes?

Answer

S3 provides multiple storage classes based on access requirements.

Examples include Standard, Intelligent-Tiering, and Glacier.

Glacier is commonly used for long-term backups and archival.

9. CloudFront

Question

What is CloudFront?

Answer

CloudFront is a Content Delivery Network service in AWS.

It improves application performance by caching content globally.

CloudFront reduces latency for users.

10. Route53 Routing Policies

Question

What routing policies are available in Route53?

Answer

Route53 supports simple, weighted, latency-based, and failover routing policies.

These policies help manage traffic efficiently and improve availability.

11. Auto Scaling Group

Question

What is Auto Scaling Group?

Answer

Auto Scaling Group automatically manages EC2 instance scaling.

It increases or decreases instances based on workload demand.

It also replaces unhealthy instances automatically.

12. CloudWatch Alarm

Question

What is CloudWatch Alarm?

Answer

CloudWatch Alarm monitors AWS metrics and triggers notifications based on thresholds.

It helps identify issues like high CPU or low disk space.

13. IAM Policy

Question

What is an IAM Policy?

Answer

IAM policy defines permissions for AWS resources.

Policies specify allowed and denied actions for users and roles.

14. Assume Role

Question

What is Assume Role in AWS?

Answer

Assume Role allows temporary access to AWS resources using IAM roles.

It improves security by avoiding permanent credentials.

15. Multi-AZ Deployment

Question

What is Multi-AZ deployment?

Answer

Multi-AZ deployment distributes resources across multiple availability zones.

It improves high availability and disaster recovery capability.

16. RDS

Question

What is RDS?

Answer

RDS stands for Relational Database Service.

It is a managed database service provided by AWS.

AWS handles backups, patching, and scaling for RDS databases.

17. AWS Shared Responsibility Model

Question

What is AWS Shared Responsibility Model?

Answer

AWS manages security of the cloud infrastructure.

Customers are responsible for security inside the cloud.

This includes IAM, application security, and operating system management.

18. Scenario Question

Question

EC2 server is not reachable. What will you check?

Answer

First, I verify instance status and health checks.

Then I check security groups and network ACL rules.

I also verify route tables and internet connectivity.

After that, I check SSH service and server resource utilization.

19. Scenario Question

Question

How do you reduce AWS infrastructure cost?

Answer

We use auto scaling and right-size EC2 instances.

Unused resources are cleaned regularly.

S3 lifecycle policies help optimize storage costs.

Monitoring helps identify unnecessary resource usage.

20. Scenario Question

Question

How do you secure AWS infrastructure?

Answer

IAM roles and least privilege access improve security.
Security groups restrict unnecessary traffic.
Encryption is enabled for storage services.
Monitoring and logging help identify suspicious activity.

21. Scenario Question

Question

How do you monitor AWS infrastructure?

Answer

We use CloudWatch dashboards, logs, and alarms.
Monitoring includes CPU, memory, network, and application health metrics.
Alerts help identify issues quickly.

22. Scenario Question

Question

How do you handle high traffic in AWS?

Answer

Auto Scaling automatically increases server capacity.
Load balancers distribute traffic across multiple instances.
CloudFront improves content delivery performance.

23. Production Scenario

Question

Database performance becomes slow. What will you do?

Answer

First, I check database CPU, memory, and storage metrics.
Then I analyze slow queries and application logs.
I also verify network latency and connection limits.
After identifying the issue, I optimize resources or scale the database.

24. Disaster Recovery

Question

How do you improve disaster recovery in AWS?

Answer

We use backups, snapshots, and multi-AZ deployments.
Important data is replicated across regions if required.
Disaster recovery plans are tested regularly.

Day 11 — Terraform + Ansible + Git + CI/CD Advanced Questions

Today focus on:

- Infrastructure as Code
- automation
- Git workflows
- CI/CD best practices
- deployment strategies
- real production scenarios

These questions are commonly asked in DevOps interviews.

Speak slowly.

Pause after every sentence.

1. Terraform

Question

What is Terraform?

Answer

Terraform is an Infrastructure as Code tool developed by HashiCorp.

It is used to automate infrastructure provisioning.

Terraform supports cloud providers like AWS, Azure, and GCP.

2. Terraform Advantages

Question

What are the advantages of Terraform?

Answer

Terraform provides automation and consistency.

Infrastructure can be version controlled using Git.

It reduces manual configuration errors.

Terraform also supports reusable modules.

3. Terraform Init

Question

What is terraform init?

Answer

terraform init initializes the Terraform working directory.

It downloads provider plugins and backend configuration.

4. Terraform Plan

Question

What is terraform plan?

Answer

terraform plan shows the infrastructure changes before applying them.

It helps validate configurations safely.

5. Terraform Apply

Question

What is terraform apply?

Answer

terraform apply creates or updates infrastructure resources.

It executes changes defined in Terraform configuration files.

6. Terraform Destroy

Question

What is terraform destroy?

Answer

terraform destroy removes infrastructure resources managed by Terraform.

It helps clean up environments when no longer required.

7. Terraform Modules

Question

What are Terraform modules?

Answer

Modules are reusable Terraform configurations.

They help standardize infrastructure creation.

Modules improve maintainability and reduce code duplication.

8. Terraform State Locking

Question

Why is state locking important?

Answer

State locking prevents multiple users from modifying Terraform state simultaneously.

It helps avoid infrastructure conflicts and corruption.

9. Ansible

Question

What is Ansible?

Answer

Ansible is a configuration management and automation tool.

It is used for server configuration, application deployment, and automation tasks.

Ansible works using SSH and does not require agents.

10. Ansible Playbook

Question

What is an Ansible playbook?

Answer

Ansible playbook is a YAML file containing automation tasks.

Playbooks define server configurations and deployment steps.

11. Git Rebase

Question

What is Git rebase?

Answer

Git rebase integrates changes from one branch into another.

It creates a cleaner commit history compared to merge.

12. Git Merge

Question

What is Git merge?

Answer

Git merge combines changes from one branch into another.

It preserves commit history and creates a merge commit.

13. Pull Request

Question

What is a Pull Request?

Answer

Pull Request is used to review and merge code changes.

Team members review code before merging into the main branch.

It improves code quality and collaboration.

14. CI vs CD

Question

What is the difference between CI and CD?

Answer

Continuous Integration focuses on automated build and testing.

Continuous Delivery focuses on automated deployment preparation.

Continuous Deployment automatically deploys changes to production.

15. Pipeline Stages

Question

What stages are commonly used in CI/CD pipelines?

Answer

Common stages include source, build, test, scan, deploy, and monitoring.

Pipelines automate the complete software delivery lifecycle.

16. Artifact Repository

Question

What is an artifact repository?

Answer

Artifact repository stores build artifacts and packages.

Examples include Nexus and JFrog Artifactory.

It helps manage application versions and dependencies.

17. DevSecOps

Question

What is DevSecOps?

Answer

DevSecOps integrates security into DevOps workflows.

Security checks are automated within CI/CD pipelines.

Vulnerability scanning and access control improve application security.

18. Trivy

Question

What is Trivy?

Answer

Trivy is a container image vulnerability scanner.

It identifies security vulnerabilities in Docker images and Kubernetes configurations.

Trivy is commonly integrated into CI/CD pipelines.

19. Scenario Question

Question

Terraform deployment failed. How will you troubleshoot?

Answer

First, I review Terraform error logs.

Then I verify provider configuration and permissions.

I also check syntax and dependency issues.

After fixing the issue, I rerun terraform plan and apply.

20. Scenario Question

Question

Git merge conflict occurred. What will you do?

Answer

First, I identify conflicting files.

Then I review and resolve code conflicts carefully.

After resolving conflicts, I test the changes and complete the merge process.

21. Scenario Question

Question

How do you secure secrets in CI/CD pipelines?

Answer

Secrets should never be hardcoded in pipelines.

We use secure secret management solutions.

Access permissions are restricted using IAM and RBAC policies.

22. Scenario Question

Question

Pipeline deployment succeeded but application failed. What will you check?

Answer

First, I check application logs and pod status.

Then I verify environment variables and configuration files.

I also test database and service connectivity.

After identifying the issue, I fix the configuration or rollback if required.

23. Production Scenario

Question

How do you manage multiple environments like dev, staging, and production?

Answer

Separate configurations and namespaces are maintained for each environment.

CI/CD pipelines handle deployments independently.

Infrastructure and access permissions are also separated for security.

24. Real-Time Experience Question

Question

What automation have you implemented in your project?

Answer

I have automated CI/CD pipelines using Jenkins and ArgoCD.

I also automated infrastructure provisioning using Terraform.

Deployment automation reduced manual effort and improved release speed.

Day 12 — Docker + Containerization + Advanced DevOps Questions

Today focus on:

- Docker internals
- containerization
- image optimization
- networking
- storage
- real production scenarios

These questions are very common for DevOps engineers.

Speak slowly.

Pause after every sentence.

1. Docker Architecture

Question

Explain Docker architecture.

Answer

Docker architecture contains Docker client, Docker daemon, and Docker registry.

Docker client sends commands to Docker daemon.

Docker daemon builds and manages containers.

Docker registry stores Docker images.

2. Docker Image

Question

What is a Docker image?

Answer

Docker image is a read-only template used to create containers.

It contains application code, libraries, and dependencies.

Images help maintain consistency across environments.

3. Docker Container

Question

What is a Docker container?

Answer

Docker container is a running instance of a Docker image.

Containers are lightweight and isolated environments.
They help applications run consistently across systems.

4. Dockerfile

Question

What is a Dockerfile?

Answer

Dockerfile is a text file containing instructions to build Docker images.
It defines application dependencies and execution steps.

5. Multi-Stage Docker Build

Question

What is multi-stage Docker build?

Answer

Multi-stage build helps reduce Docker image size.
Separate stages are used for build and runtime environments.
Only required files are copied into the final image.

6. Docker Volumes

Question

What are Docker volumes?

Answer

Docker volumes provide persistent storage for containers.
Data remains available even if containers are removed.
Volumes help manage application data efficiently.

7. Docker Networking

Question

What types of Docker networks are available?

Answer

Docker supports bridge, host, overlay, and none networks.
Bridge network is commonly used for container communication on a single host.
Overlay network is used in container orchestration platforms.

8. Docker Registry

Question

What is a Docker registry?

Answer

Docker registry stores and distributes Docker images.
Docker Hub and AWS ECR are commonly used registries.

9. Docker Compose

Question

What is Docker Compose?

Answer

Docker Compose is used to manage multi-container applications.
Services are defined in a YAML configuration file.
It simplifies local development and testing.

10. Docker vs Virtual Machine

Question

What is the difference between Docker and Virtual Machine?

Answer

Docker containers share the host operating system kernel.
Virtual machines contain separate operating systems.
Containers are lightweight and faster compared to virtual machines.

11. Containerization Benefits

Question

What are the benefits of containerization?

Answer

Containerization improves portability and consistency.
Applications run the same across different environments.
Containers also improve scalability and deployment speed.

12. Docker Logs

Question

How do you check container logs?

Answer

We use docker logs command to check container logs.
Logs help identify application and startup issues.

13. Docker Restart Policy

Question

What are Docker restart policies?

Answer

Restart policies control container restart behavior.

Common policies include always, on-failure, and unless-stopped.

14. Docker Resource Limits

Question

Can we limit container resources in Docker?

Answer

Yes.

Docker supports CPU and memory limits for containers.

Resource limits help prevent resource exhaustion on servers.

15. Docker Security

Question

How do you secure Docker containers?

Answer

We use minimal base images and vulnerability scanning.

Containers run with restricted permissions.

Secrets are managed securely and images are updated regularly.

16. Scenario Question

Question

Container is exiting immediately. How will you troubleshoot?

Answer

First, I check container logs using `docker logs` command.

Then I verify application startup configuration.

I also check environment variables and dependencies.

After identifying the issue, I rebuild and restart the container.

17. Scenario Question

Question

Docker image size is very large. How will you optimize it?

Answer

We use lightweight base images like Alpine Linux.

Multi-stage builds help remove unnecessary build dependencies.

Unused packages and cache files are removed from images.

18. Scenario Question

Question

Container cannot connect to database. What will you check?

Answer

First, I verify database service availability.

Then I check network connectivity and environment variables.

I also verify firewall rules and port configurations.

Finally, I test connectivity from inside the container.

19. Scenario Question

Question

Docker daemon is not working. What will you do?

Answer

First, I check Docker service status using `systemctl`.

Then I review Docker daemon logs for errors.

I also verify disk space and server resources.

After fixing the issue, I restart Docker service.

20. Real-Time Experience Question

Question

How are you using Docker in your project?

Answer

Developers create application code and push it to Git repositories.

Jenkins pipelines build Docker images automatically.

Images are pushed to container registries.

Kubernetes deploys the containers into EKS clusters.

21. Production Scenario

Question

How do you handle container failures in production?

Answer

First, I identify the issue using logs and monitoring tools.

Then I verify application health and resource usage.

I restart or redeploy containers if required.

Root cause analysis helps prevent future failures.

22. Advanced Question

Question

What is the difference between `CMD` and `ENTRYPOINT` in Docker?

Answer

CMD provides default commands for containers.

ENTRYPOINT defines the main executable for the container.

ENTRYPOINT is generally not overridden during runtime.

23. Advanced Question

Question

What is the difference between COPY and ADD in Dockerfile?

Answer

COPY is used to copy files from local system into Docker image.

ADD provides additional features like extracting archives and downloading URLs.

COPY is generally preferred for simplicity and security.

24. Production Best Practice

Question

What best practices do you follow while creating Docker images?

Answer

We use lightweight and secure base images.

Multi-stage builds help reduce image size.

Unnecessary packages are removed.

Vulnerability scanning is integrated into CI/CD pipelines.

Day 13 — ArgoCD + GitOps + Helm + Deployment Strategy Questions

Today focus on:

- ArgoCD
- GitOps workflow
- Helm deployments
- deployment strategies
- rollback handling
- production automation

These questions are very common for Kubernetes DevOps roles.

Speak slowly.

Pause after every sentence.

1. ArgoCD

Question

What is ArgoCD?

Answer

ArgoCD is a GitOps continuous delivery tool for Kubernetes.

It automatically deploys applications from Git repositories.

ArgoCD continuously monitors Git and syncs changes to Kubernetes clusters.

2. GitOps

Question

What is GitOps?

Answer

GitOps is a deployment approach where Git acts as the single source of truth.

Infrastructure and deployment configurations are stored in Git repositories.

Changes are automatically deployed using tools like ArgoCD.

3. Benefits of GitOps

Question

What are the benefits of GitOps?

Answer

GitOps improves automation and consistency.

It provides version control for infrastructure changes.

Rollback and auditing become easier using Git history.

4. ArgoCD Sync

Question

What is Sync in ArgoCD?

Answer

Sync means applying Git repository changes to the Kubernetes cluster.

ArgoCD compares desired state from Git with the actual cluster state.

During sync, missing or updated resources are deployed automatically.

5. Auto Sync

Question

What is Auto Sync in ArgoCD?

Answer

Auto Sync automatically deploys changes whenever Git repository updates occur.

It reduces manual deployment effort and improves automation.

6. Drift Detection

Question

What is Drift Detection in ArgoCD?

Answer

Drift detection identifies differences between Git configuration and cluster state.

ArgoCD continuously monitors for configuration drift.

7. Helm

Question

What is Helm?

Answer

Helm is a package manager for Kubernetes.

It simplifies application deployment and management.

Helm uses charts to manage Kubernetes resources.

8. Helm Chart

Question

What is a Helm chart?

Answer

Helm chart is a collection of Kubernetes YAML templates.

It contains deployment files, service definitions, and configurable values.

Helm charts help standardize deployments.

9. values.yaml

Question

What is values.yaml in Helm?

Answer

values.yaml stores configurable values for Helm charts.

It helps customize deployments for different environments.

10. Helm Upgrade

Question

How do you upgrade applications using Helm?

Answer

Helm upgrade command is used to update application deployments.

It applies updated chart configurations to Kubernetes clusters.

11. Helm Rollback

Question

How do you rollback in Helm?

Answer

Helm rollback command restores the previous stable release version.

It helps recover applications quickly during deployment failures.

12. Deployment Strategy

Question

What deployment strategies have you used?

Answer

I have worked with rolling update, blue-green, and canary deployment strategies. These strategies help reduce downtime and deployment risk.

13. Rolling Update

Question

What is Rolling Update?

Answer

Rolling update replaces old application pods gradually with new pods. It helps minimize downtime during deployments.

14. Blue-Green Deployment

Question

What is Blue-Green deployment?

Answer

Blue-Green deployment uses two identical environments. One environment runs the current version and another runs the new version. Traffic is switched after successful validation.

15. Canary Deployment

Question

What is Canary deployment?

Answer

Canary deployment releases the new version to a small set of users initially. If the application works correctly, deployment continues gradually. This reduces deployment risk in production.

16. Rollback Strategy

Question

What is your rollback strategy?

Answer

Before deployment, backup and rollback plans are prepared. If deployment fails, we rollback to the previous stable version immediately. Monitoring and validation help ensure application stability after rollback.

17. Argo Rollouts

Question

What is Argo Rollouts?

Answer

Argo Rollouts is a Kubernetes deployment controller.

It supports advanced deployment strategies like canary and blue-green deployments.

It improves deployment control and application reliability.

18. Scenario Question

Question

ArgoCD application is OutOfSync. What will you do?

Answer

First, I compare Git configuration with cluster resources.

Then I identify manual changes or deployment differences.

After that, I perform sync operation to restore the desired state.

19. Scenario Question

Question

Helm deployment failed. How will you troubleshoot?

Answer

First, I check Helm release status and error logs.

Then I validate chart syntax and Kubernetes events.

I also verify values.yaml configuration and resource availability.

After fixing the issue, I redeploy the application.

20. Scenario Question

Question

How do you manage multiple environments using Helm?

Answer

Separate values files are maintained for development, staging, and production.

Environment-specific configurations are managed through Helm values.

This improves deployment consistency across environments.

21. Production Experience Question

Question

How are you using ArgoCD in your project?

Answer

We use ArgoCD for GitOps-based continuous deployment.

Kubernetes manifests and Helm charts are stored in Git repositories.
ArgoCD automatically syncs and deploys applications to EKS clusters.

22. Production Experience Question

Question

How do you ensure deployment safety?

Answer

We use automated testing and monitoring before production deployment.
Rolling updates and health checks reduce downtime.
Rollback plans are prepared before every release.

23. Security Question

Question

How do you secure GitOps workflows?

Answer

Access control is managed using RBAC and IAM policies.
Sensitive data is stored securely using secrets management solutions.
Git repository access is restricted and audited regularly.

24. Best Practice Question

Question

What best practices do you follow for Kubernetes deployments?

Answer

We use version-controlled deployment configurations.
Resource limits and health checks are configured properly.
Monitoring and rollback strategies are implemented.
Automation improves deployment consistency and reliability.

Day 14 — Advanced Linux + Shell Scripting + Monitoring Interview Questions

Today focus on:

- Linux administration
- shell scripting
- monitoring
- troubleshooting
- process management
- production debugging

These questions are very common in DevOps interviews.

Speak slowly.

Pause after every sentence.

1. Linux File Permissions

Question

What are Linux file permissions?

Answer

Linux permissions control file access for users and groups.

Permissions include read, write, and execute access.

chmod command is used to modify permissions.

2. chmod Command

Question

What is chmod 755?

Answer

chmod 755 gives full permissions to the owner.

Group and others get read and execute permissions only.

3. chown Command

Question

What is chown command?

Answer

chown is used to change file ownership in Linux.

It assigns files to specific users and groups.

4. grep Command

Question

What is grep used for?

Answer

grep searches specific text patterns inside files.

It is commonly used for log analysis and troubleshooting.

5. find Command

Question

What is the find command?

Answer

find command searches files and directories in Linux.

It can search based on name, size, and permissions.

6. awk Command

Question

What is awk command?

Answer

awk is used for text processing and data extraction.

It is commonly used for parsing logs and reports.

7. sed Command

Question

What is sed command?

Answer

sed is a stream editor used for modifying text files.

It supports search and replace operations.

8. Soft Link vs Hard Link

Question

What is the difference between soft link and hard link?

Answer

Soft link points to the original file path.

Hard link points directly to the file inode.

Soft links can work across file systems, but hard links cannot.

9. Process Management

Question

How do you check running processes in Linux?

Answer

We use ps and top commands to monitor running processes.

These commands help identify resource-consuming applications.

10. Kill Command

Question

What is kill command?

Answer

kill command is used to terminate processes in Linux.

It sends signals to running processes.

11. Crontab

Question

What is crontab?

Answer

crontab is used to schedule recurring tasks in Linux.
It automates jobs like backups and cleanup scripts.

12. Shell Scripting

Question

What is shell scripting?

Answer

Shell scripting automates Linux tasks using scripts.
It helps reduce manual work and improve efficiency.

13. Variables in Shell Script

Question

How do you define variables in shell scripting?

Answer

Variables are assigned using equal sign.

Example:

```
name=Vipin
```

Variables help store reusable values.

14. Monitoring Tools

Question

What monitoring tools have you used?

Answer

I have worked with Prometheus, Grafana, and CloudWatch.
These tools help monitor infrastructure and application health.

15. Prometheus

Question

How does Prometheus work?

Answer

Prometheus collects metrics from applications and servers.
It uses pull-based monitoring.
Metrics are stored and visualized using Grafana dashboards.

16. Grafana

Question

What is Grafana?

Answer

Grafana is a visualization tool for monitoring dashboards.

It displays metrics using graphs and alerts.

17. Alerting

Question

How do you configure monitoring alerts?

Answer

Alerts are configured based on CPU, memory, disk, and application health thresholds.

Monitoring tools send notifications during failures.

18. Scenario Question

Question

Linux server CPU usage is high. How will you troubleshoot?

Answer

First, I check running processes using top command.

Then I identify high CPU-consuming applications.

I review logs and traffic patterns.

After identifying the issue, I restart services or optimize resources.

19. Scenario Question

Question

Disk space is full. What will you do?

Answer

First, I check disk usage using df -h command.

Then I identify large files using du command.

I remove unnecessary logs and temporary files.

If required, I increase storage capacity.

20. Scenario Question

Question

Application process is not running. What will you check?

Answer

First, I verify process status using ps command.

Then I check application logs for errors.

I also verify port availability and resource usage.

After identifying the issue, I restart the application service.

21. Scenario Question

Question

How do you automate repetitive tasks in Linux?

Answer

Shell scripting and crontab are commonly used for automation.

Scripts help automate backups, cleanup, monitoring, and deployment tasks.

22. Production Experience Question**Question**

How do you use Linux in your project?

Answer

Linux servers are used for Jenkins, Docker, and Kubernetes environments.

I perform troubleshooting, monitoring, automation, and deployment activities on Linux systems.

23. Security Question**Question**

How do you improve Linux server security?

Answer

We restrict unnecessary access and services.

SSH access is secured using keys and permissions.

Regular patching and monitoring improve server security.

24. Real-Time Troubleshooting**Question**

How do you analyze logs during production issues?

Answer

First, I identify the affected application or service.

Then I filter logs using grep and monitoring tools.

I correlate logs with monitoring alerts and deployment changes.

After identifying the root cause, I fix the issue and monitor stability.

Day 15 — Kubernetes Troubleshooting Master Questions

Today focus on:

- real production troubleshooting
- Kubernetes debugging
- networking issues
- scaling problems
- storage failures
- cluster recovery

These questions are very important for experienced DevOps engineers.

Speak slowly.
Pause after every sentence.

1. Pod Pending State

Question

A pod is stuck in Pending state. How will you troubleshoot?

Answer

First, I check pod events using `kubectl describe pod` command.

Then I verify node resources like CPU and memory.

I also check taints, tolerations, and node selectors.

If storage is used, I verify PVC binding issues.

After identifying the issue, I fix the scheduling problem.

2. CrashLoopBackOff

Question

How do you troubleshoot CrashLoopBackOff?

Answer

First, I check pod logs using `kubectl logs` command.

Then I verify application startup configuration.

I also check environment variables, secrets, and database connectivity.

After identifying the root cause, I update the configuration and restart the deployment.

3. ImagePullBackOff

Question

What causes ImagePullBackOff?

Answer

ImagePullBackOff occurs when Kubernetes cannot pull container images.

Common reasons include incorrect image name, authentication failure, or missing image tags.

I verify registry access and deployment configuration.

4. OOMKilled

Question

What is OOMKilled and how do you fix it?

Answer

OOMKilled happens when a container exceeds memory limits.

Kubernetes terminates the container to protect node stability.

I analyze memory usage and increase limits if required.

Application optimization may also be needed.

5. Node NotReady

Question

How do you troubleshoot a node in NotReady state?

Answer

First, I check node status and events.

Then I verify kubelet service and node connectivity.

I also check CPU, memory, and disk utilization.

After identifying the issue, I restart services or recover the node.

6. Kubernetes DNS Issue

Question

How do you troubleshoot DNS issues in Kubernetes?

Answer

First, I verify CoreDNS pod status.

Then I test DNS resolution from inside application pods.

I also check service names and namespace configuration.

After identifying the issue, I fix DNS or networking configuration.

7. Service Not Accessible

Question

Application pods are running but service is not accessible. What will you check?

Answer

First, I verify service configuration and selectors.

Then I check pod labels and endpoints.

I also verify ingress and network connectivity.

Finally, I test application ports and logs.

8. Ingress Issue

Question

How do you troubleshoot ingress issues?

Answer

First, I check ingress resource configuration.

Then I verify ingress controller status.

I also check DNS mapping and SSL configuration.

After that, I review application service connectivity.

9. Persistent Volume Issue

Question

PVC is not binding. What will you do?

Answer

First, I check PVC and PV status.

Then I verify storage class configuration.

I also check available storage capacity and access modes.

After fixing the issue, I verify successful binding.

10. High CPU Usage

Question

Pods are consuming very high CPU. How will you handle it?

Answer

First, I check resource metrics and monitoring dashboards.

Then I identify high CPU-consuming applications.

I increase pod replicas using HPA if traffic is high.

Application optimization may also be required.

11. Deployment Failure

Question

Deployment failed after release. What will you do?

Answer

First, I check deployment events and pod logs.

Then I verify image versions and configuration changes.

If the issue is critical, I perform rollback to the previous stable version.

After fixing the issue, I redeploy and monitor the application.

12. Kubernetes Networking

Question

How do pods communicate across nodes?

Answer

Kubernetes networking allows pod-to-pod communication across nodes.

Each pod receives a unique IP address.

Container Network Interface plugins handle networking functionality.

13. CNI

Question

What is CNI in Kubernetes?

Answer

CNI stands for Container Network Interface.

It manages networking between Kubernetes pods.

Examples include Calico, Flannel, and Cilium.

14. Cluster Upgrade

Question

How do you upgrade Kubernetes clusters safely?

Answer

First, I review compatibility and backup configurations.

Then I upgrade the control plane and worker nodes gradually.

Application health and monitoring are verified during the process.

Rollback planning is also prepared.

15. etcd Backup

Question

Why is etcd backup important?

Answer

etcd stores Kubernetes cluster state and configuration.

Backup helps recover the cluster during failures.

Regular etcd backup is important for disaster recovery.

16. Resource Quota

Question

What is ResourceQuota in Kubernetes?

Answer

ResourceQuota limits resource consumption within namespaces.

It helps control CPU, memory, and storage usage.

17. LimitRange

Question

What is LimitRange?

Answer

LimitRange defines default resource limits and requests for containers in a namespace.

It improves resource management and cluster stability.

18. Scenario Question

Question

Pods are restarting frequently after deployment. What will you check?

Answer

First, I check pod logs and events.

Then I verify readiness and liveness probes.

I also check environment variables and database connectivity.

Resource limits and application errors are also analyzed.

19. Scenario Question

Question

Cluster performance is slow. How will you troubleshoot?

Answer

First, I check node resource utilization.

Then I analyze pod metrics and monitoring dashboards.

I also verify storage, networking, and application performance.

After identifying bottlenecks, I optimize resources or scale infrastructure.

20. Scenario Question

Question

How do you handle production incidents in Kubernetes?

Answer

First, I identify the affected application or service.

Then I check logs, metrics, and recent deployment changes.

I coordinate with teams and restore services quickly.

Root cause analysis is performed after resolution.

21. Real-Time Experience Question

Question

What Kubernetes troubleshooting have you done in production?

Answer

I have handled pod crashes, deployment failures, DNS issues, and resource utilization problems.

I troubleshoot logs, networking, and application configurations regularly.

Monitoring and rollback strategies help maintain production stability.

22. Security Question

Question

How do you improve Kubernetes security?

Answer

We use RBAC for access control.

Secrets are used for sensitive data management.

Network policies restrict unnecessary communication.

Image scanning and regular patching improve cluster security.

Day 16 — Real-Time DevOps Production Scenario Questions

Today focus on:

- real production incidents
- troubleshooting approach
- communication during outages
- deployment recovery
- monitoring and debugging

These are high-value interview questions for experienced DevOps engineers.

Speak slowly.

Pause after every sentence.

1. Production Deployment Failure

Question

A production deployment failed during release. What will you do?

Answer

First, I check deployment logs and monitoring alerts.

Then I identify whether the issue is related to application code, infrastructure, or configuration.

If the issue is critical, I rollback to the previous stable version.

After restoring the service, I analyze the root cause and redeploy after fixing the issue.

2. Application Downtime

Question

Application is completely down in production. How will you troubleshoot?

Answer

First, I verify whether pods or servers are running properly.

Then I check load balancer, ingress, and DNS configuration.

I also analyze logs and monitoring dashboards.

After identifying the root cause, I restore application availability as quickly as possible.

3. Database Connection Failure

Question

Application cannot connect to database. What will you check?

Answer

First, I verify database availability and health status.

Then I check connection strings, secrets, and environment variables.

I also verify firewall rules and network connectivity.

Finally, I analyze application logs and database logs.

4. Jenkins Build Failure

Question

Jenkins build suddenly started failing. How will you troubleshoot?

Answer

First, I check Jenkins console output logs.

Then I identify which stage is failing.

I verify Git access, build dependencies, and Docker configuration.

After fixing the issue, I rerun the pipeline and validate deployment.

5. Kubernetes Cluster Issue

Question

Multiple pods are failing across the cluster. What will you check?

Answer

First, I check node health and cluster events.

Then I verify resource utilization and networking status.

I also check whether recent deployments or infrastructure changes caused the issue.

After identifying the root cause, I restore cluster stability.

6. Memory Leak

Question

How do you identify memory leak issues?

Answer

First, I monitor memory usage trends using monitoring tools.

Then I analyze application logs and container metrics.

If memory usage continuously increases without release, it may indicate a memory leak.

I coordinate with developers for application optimization.

7. High Traffic Incident

Question

Traffic suddenly increased heavily. How will you handle it?

Answer

First, I monitor infrastructure and application metrics.

Then I verify auto scaling and load balancer performance.

Kubernetes HPA and AWS Auto Scaling help handle traffic automatically.

I continue monitoring system stability during the traffic spike.

8. SSL Certificate Expired

Question

Production SSL certificate expired. What will you do?

Answer

First, I verify certificate expiration details.

Then I renew or replace the SSL certificate.

After that, I update ingress or load balancer configuration.

Finally, I test HTTPS connectivity and monitor application access.

9. Rollback Scenario

Question

When will you perform rollback?

Answer

Rollback is performed when deployment causes critical production issues.

Examples include application crashes, high error rates, or major functionality failures.

Rollback helps restore service stability quickly.

10. Root Cause Analysis

Question

What is Root Cause Analysis?

Answer

Root Cause Analysis identifies the actual reason behind an incident.

It helps prevent repeated failures in production.

Logs, monitoring, and deployment history are analyzed during RCA.

11. Incident Communication

Question

How do you communicate during production incidents?

Answer

I provide regular updates to teams and stakeholders.

Communication includes issue impact, troubleshooting progress, and estimated resolution time.

Clear communication reduces confusion during incidents.

12. Monitoring Dashboard

Question

What metrics do you monitor in production?

Answer

We monitor CPU, memory, disk, network, and application response times.

Pod health and deployment status are also monitored.

Alerts are configured for critical thresholds.

13. Log Analysis

Question

How do you analyze production logs?

Answer

First, I identify the affected application or service.

Then I filter logs using grep and monitoring tools.

I correlate logs with alerts and deployment changes.

This helps identify the root cause quickly.

14. Kubernetes Restart Issue

Question

Pods keep restarting after deployment. What will you check?

Answer

First, I check pod logs and events.

Then I verify readiness and liveness probes.

I also check environment variables and application startup configuration.

Resource limits and dependency issues are also verified.

15. Disk Space Full

Question

Production server disk is full. What will you do?

Answer

First, I check disk usage using df -h command.

Then I identify large files and logs using du command.

I clean unnecessary files and rotate logs if required.

Storage expansion may also be needed.

16. Deployment Strategy

Question

Which deployment strategy do you prefer and why?

Answer

I prefer rolling updates for regular deployments because they reduce downtime.

For critical applications, canary or blue-green deployment provides safer releases.

Deployment strategy depends on business requirements and risk level.

17. Production Best Practice

Question

What best practices do you follow before production deployment?

Answer

We validate application testing and deployment configuration.
Monitoring and rollback plans are prepared before deployment.
Backups are verified for critical systems.
Deployment is monitored carefully after release.

18. Security Incident

Question

How do you handle security incidents?

Answer

First, I identify affected systems and isolate the issue.
Then I review logs and access history.
Access permissions and credentials are updated if required.
Monitoring and security improvements are implemented after resolution.

19. Team Coordination

Question

How do you coordinate with teams during incidents?

Answer

I communicate regularly with developers, QA, and operations teams.
I share troubleshooting findings and deployment updates.
Collaboration helps resolve incidents faster.

20. Real-Time Experience Question

Question

Tell me about a critical production issue you handled.

Answer

One time, application pods started crashing after deployment.
I checked logs and identified incorrect database configuration.
I performed rollback immediately and restored application stability.
After fixing the configuration, deployment was completed successfully.

Day 17 — Advanced Networking + Load Balancer + DNS + Security Questions

Today focus on:

- networking fundamentals
- Kubernetes networking
- AWS networking
- DNS

- SSL
- security concepts

These questions are very common in DevOps interviews.

Speak slowly.

Pause after every sentence.

1. OSI Model

Question

What is the OSI model?

Answer

OSI model is a networking framework containing seven layers.

It helps understand network communication processes.

Layers include physical, data link, network, transport, session, presentation, and application.

2. TCP/IP

Question

What is TCP/IP?

Answer

TCP/IP is a networking protocol suite used for internet communication.

TCP provides reliable communication.

IP handles addressing and routing.

3. IP Address

Question

What is an IP address?

Answer

IP address uniquely identifies devices in a network.

It helps systems communicate with each other.

4. Public IP vs Private IP

Question

What is the difference between public IP and private IP?

Answer

Public IP is accessible over the internet.

Private IP is used inside internal networks only.

Private IP improves security and internal communication.

5. DNS

Question

What is DNS?

Answer

DNS stands for Domain Name System.

It converts domain names into IP addresses.

DNS helps users access applications using easy-to-remember names.

6. Load Balancer

Question

What is a Load Balancer?

Answer

Load balancer distributes incoming traffic across multiple servers or pods.

It improves application availability and performance.

Load balancing helps avoid server overload.

7. Application Load Balancer

Question

What is an Application Load Balancer?

Answer

Application Load Balancer works at Layer 7 of the OSI model.

It supports HTTP and HTTPS traffic routing.

ALB supports path-based and host-based routing.

8. Network Load Balancer

Question

What is a Network Load Balancer?

Answer

Network Load Balancer works at Layer 4.

It handles TCP and UDP traffic with very high performance.

9. Reverse Proxy

Question

What is a Reverse Proxy?

Answer

Reverse proxy receives client requests and forwards them to backend servers.

It improves security, load balancing, and performance.

10. SSL Certificate

Question

What is an SSL certificate?

Answer

SSL certificate encrypts communication between client and server.

It improves application security using HTTPS protocol.

11. HTTPS

Question

What is HTTPS?

Answer

HTTPS is a secure version of HTTP.

It uses SSL or TLS encryption for secure communication.

12. Firewall

Question

What is a Firewall?

Answer

Firewall controls network traffic based on security rules.

It helps prevent unauthorized access.

13. Port Numbers

Question

What are commonly used ports?

Answer

Port 22 is used for SSH.

Port 80 is used for HTTP.

Port 443 is used for HTTPS.

Port 3306 is commonly used for MySQL.

14. NAT Gateway

Question

What is NAT Gateway?

Answer

NAT Gateway allows private subnet resources to access the internet securely.

It prevents direct inbound internet access to private resources.

15. VPN

Question

What is VPN?

Answer

VPN stands for Virtual Private Network.

It provides secure communication between networks over the internet.

16. Kubernetes Networking**Question**

How does Kubernetes networking work?

Answer

Kubernetes assigns a unique IP address to each pod.

Pods communicate directly without NAT.

CNI plugins handle pod networking functionality.

17. Service Mesh**Question**

What is Service Mesh?

Answer

Service mesh manages communication between microservices.

It provides traffic management, security, and observability.

Istio is a commonly used service mesh tool.

18. Ingress**Question**

What is Ingress?

Answer

Ingress manages external access to Kubernetes applications.

It provides routing, SSL termination, and load balancing.

19. Network Policy**Question**

What is Network Policy?

Answer

Network Policy controls traffic flow between Kubernetes pods.

It improves security by restricting unauthorized communication.

20. Scenario Question**Question**

Application is not reachable through domain name. What will you check?

Answer

First, I verify DNS records and domain configuration.
Then I check load balancer and ingress status.
I also verify SSL certificates and application services.
Finally, I test network connectivity and application ports.

21. Scenario Question

Question

How do you troubleshoot SSL issues?

Answer

First, I verify SSL certificate validity and expiration.
Then I check ingress or load balancer SSL configuration.
I also verify DNS mapping and HTTPS connectivity.

22. Scenario Question

Question

Users are experiencing slow application response. What will you check?

Answer

First, I monitor CPU, memory, and network metrics.
Then I verify load balancer health and backend services.
I also analyze database performance and application logs.
Scaling or optimization may be required.

23. Security Best Practice

Question

How do you secure Kubernetes networking?

Answer

Network policies restrict unnecessary pod communication.
Ingress rules control external traffic access.
TLS encryption improves communication security.

24. Real-Time Experience Question

Question

How are networking components used in your project?

Answer

We use AWS load balancers, ingress controllers, Route53, and Kubernetes services.
These components help manage traffic routing and secure application access.

1. Dockerfile Interview Question

Question

Write a Dockerfile for a Node.js application.

Answer

```
FROM node:18
```

```
WORKDIR /app
```

```
COPY package*.json ./
```

```
RUN npm install
```

```
COPY . .
```

```
EXPOSE 3000
```

```
CMD ["npm", "start"]
```

Explanation

First, we use the Node.js base image.

WORKDIR creates the application directory.

COPY copies package files into the container.

RUN npm install installs dependencies.

Then application code is copied.

EXPOSE defines the application port.

CMD starts the application.

2. Multi-Stage Dockerfile

Question

Write a multi-stage Dockerfile.

Answer

```
# Build Stage
```

```
FROM node:18 AS builder
```

```
WORKDIR /app
```

```
COPY package*.json ./
```

```
RUN npm install
```

```
COPY . .
```

```
RUN npm run build
```

```
# Runtime Stage
```

```
FROM nginx:alpine
```

```
COPY --from=builder /app/build /usr/share/nginx/html
```

```
EXPOSE 80
```

```
CMD ["nginx", "-g", "daemon off;"]
```

Interview Explanation

Multi-stage build helps reduce Docker image size.

The first stage builds the application.

The second stage contains only runtime files.

This improves security and performance.

3. Terraform EC2 Instance

Question

Write Terraform code to create an EC2 instance.

Answer

```
provider "aws" {  
  region = "ap-south-1"  
}  
  
resource "aws_instance" "myec2" {  
  ami      = "ami-0f58b397bc5c1f2e8"  
  instance_type = "t2.micro"  
  
  tags = {  
    Name = "DevOps-Server"  
  }  
}
```

Interview Explanation

Provider block defines the AWS region.

aws_instance resource creates the EC2 instance.

AMI defines the operating system image.

Instance type defines CPU and memory configuration.

Tags help identify resources.

4. Terraform Commands

Question

What commands are used in Terraform workflow?

Answer

terraform init

terraform plan

terraform apply

terraform destroy

Interview Explanation

terraform init initializes the project.

terraform plan previews infrastructure changes.

terraform apply creates resources.

terraform destroy removes infrastructure.

5. Kubernetes Deployment YAML

Question

Write a Kubernetes deployment YAML.

Answer

apiVersion: apps/v1

kind: Deployment

metadata:

name: nginx-deployment

spec:

replicas: 2

selector:

matchLabels:

app: nginx

template:

```
metadata:
  labels:
    app: nginx

spec:
  containers:
    - name: nginx
      image: nginx:latest

  ports:
    - containerPort: 80
```

Interview Explanation

Deployment manages application pods.

Replicas define the number of pod copies.

Selector matches pod labels.

Container section defines image and port configuration.

6. Kubernetes Service YAML

Question

Write a Kubernetes service YAML.

Answer

```
apiVersion: v1
```

```
kind: Service
```

```
metadata:
```

```
  name: nginx-service
```

```
spec:
```

```
  selector:
```

```
    app: nginx
```

```
  ports:
```

```
    - protocol: TCP
```

```
      port: 80
```

```
      targetPort: 80
```

```
  type: LoadBalancer
```

Interview Explanation

Service exposes application pods.

Selector connects service with pods.

Port defines external access port.

LoadBalancer type exposes application externally.

7. Jenkins Pipeline

Question

Write a simple Jenkins pipeline.

Answer

```
pipeline {  
    agent any  
  
    stages {  
  
        stage('Build') {  
            steps {  
                echo 'Building Application'  
            }  
        }  
  
        stage('Test') {  
            steps {  
                echo 'Running Tests'  
            }  
        }  
  
        stage('Deploy') {  
            steps {  
                echo 'Deploying Application'  
            }  
        }  
    }  
}
```

Interview Explanation

Pipeline contains multiple stages.

Build stage compiles the application.

Test stage validates the application.

Deploy stage deploys the application.

8. Kubernetes Commands

Question

What commands do you commonly use in Kubernetes?

Answer

kubectl get pods

kubectl describe pod pod-name

kubectl logs pod-name

kubectl get svc

kubectl get nodes

kubectl apply -f deployment.yaml

Interview Explanation

kubectl get pods shows pod status.

describe command provides detailed information.

logs command helps troubleshoot issues.

apply command deploys Kubernetes resources.

9. Docker Commands

Question

What Docker commands do you use daily?

Answer

docker build -t myapp .

docker images

docker ps

docker logs container-id

docker exec -it container-id bash

docker stop container-id

Interview Explanation

docker build creates images.

docker ps shows running containers.

docker logs checks container logs.

docker exec accesses container terminal.

10. Real Interview Coding Question

Question

Write deployment and service for NGINX.

Answer

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: nginx-deployment
```

```
spec:
```

```
  replicas: 3
```

```
  selector:
```

```
    matchLabels:
```

```
      app: nginx
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        app: nginx
```

```
    spec:
```

```
      containers:
```

```
        - name: nginx
```

```
          image: nginx:latest
```

```
      ports:
```

```
        - containerPort: 80
```

```
---
```

```
apiVersion: v1
```

```
kind: Service
```

```
metadata:
```

```
  name: nginx-service
```

```
spec:
```

selector:
app: nginx

ports:
- port: 80
targetPort: 80

type: LoadBalancer

Day 18 — Advanced Live Coding + YAML + DevOps Hands-On Questions

Today focus on:

- live coding
- Kubernetes YAML
- Helm
- Ingress
- ConfigMap
- Secret
- Terraform advanced code

These are very common practical interview questions.

Speak slowly.

Pause after every sentence.

1. Kubernetes ConfigMap

Question

Write a ConfigMap YAML.

Answer

apiVersion: v1
kind: ConfigMap

metadata:
name: app-config

data:
APP_ENV: production
LOG_LEVEL: info

Interview Explanation

ConfigMap stores non-sensitive configuration data.

Applications can access these values as environment variables.

2. Kubernetes Secret

Question

Write a Secret YAML.

Answer

```
apiVersion: v1
```

```
kind: Secret
```

```
metadata:
```

```
  name: db-secret
```

```
type: Opaque
```

```
data:
```

```
  username: YWRtaW4=
```

```
  password: YWRtaW4xMjM=
```

Interview Explanation

Secret stores sensitive information like passwords and API keys.

Values are stored in base64 encoded format.

3. Deployment with ConfigMap

Question

How do you use ConfigMap in Deployment?

Answer

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: nginx-deployment
```

```
spec:
```

```
  replicas: 2
```

```
  selector:
```

```
    matchLabels:
```

```
      app: nginx
```

```
template:
  metadata:
    labels:
      app: nginx

  spec:
    containers:
      - name: nginx
        image: nginx

        env:
          - name: APP_ENV
            valueFrom:
              configMapKeyRef:
                name: app-config
                key: APP_ENV
```

Interview Explanation

valueFrom is used to read ConfigMap values.

Environment variables are injected into containers dynamically.

4. Deployment with Secret

Question

How do you use Secret in Deployment?

Answer

```
env:
  - name: DB_USERNAME

valueFrom:
  secretKeyRef:
    name: db-secret
    key: username
```

5. Ingress YAML

Question

Write an Ingress YAML.

Answer

```
apiVersion: networking.k8s.io/v1
```

kind: Ingress

metadata:

name: nginx-ingress

spec:

rules:

- host: myapp.example.com

http:

paths:

- path: /

pathType: Prefix

backend:

service:

name: nginx-service

port:

number: 80

Interview Explanation

Ingress manages external HTTP and HTTPS traffic.

Host-based routing is configured using domain names.

6. Persistent Volume

Question

Write a Persistent Volume YAML.

Answer

apiVersion: v1

kind: PersistentVolume

metadata:

name: my-pv

spec:

capacity:

storage: 1Gi

accessModes:

- ReadWriteOnce

hostPath:

path: "/data"

7. Persistent Volume Claim

Question

Write a PVC YAML.

Answer

apiVersion: v1

kind: PersistentVolumeClaim

metadata:

name: my-pvc

spec:

accessModes:

- ReadWriteOnce

resources:

requests:

storage: 1Gi

8. HPA YAML

Question

Write HPA YAML.

Answer

apiVersion: autoscaling/v2

kind: HorizontalPodAutoscaler

metadata:

name: nginx-hpa

spec:

scaleTargetRef:

apiVersion: apps/v1
kind: Deployment
name: nginx-deployment

minReplicas: 2
maxReplicas: 5

metrics:
- type: Resource

resource:
name: cpu

target:
type: Utilization
averageUtilization: 70

Interview Explanation

HPA automatically scales pods based on CPU or memory usage.

9. Terraform Variables

Question

Write Terraform variable example.

Answer

```
variable "instance_type" {  
  default = "t2.micro"  
}
```

10. Terraform Output

Question

Write Terraform output block.

Answer

```
output "instance_ip" {  
  value = aws_instance.myec2.public_ip  
}
```

11. Terraform Security Group

Question

Write Terraform code for Security Group.

Answer

```
resource "aws_security_group" "my-sg" {
```

```
  name = "web-sg"
```

```
  ingress {
    from_port = 22
    to_port   = 22
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
```

```
  ingress {
    from_port = 80
    to_port   = 80
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
}
```

12. Terraform VPC

Question

Write basic Terraform VPC code.

Answer

```
resource "aws_vpc" "main" {
```

```
  cidr_block = "10.0.0.0/16"
```

```
  tags = {
    Name = "devops-vpc"
  }
}
```

13. Helm values.yaml

Question

Write Helm values.yaml example.

Answer

replicaCount: 2

image:

repository: nginx

tag: latest

service:

type: LoadBalancer

port: 80

14. Helm Template

Question

Write Helm deployment template snippet.

Answer

```
replicas: {{ .Values.replicaCount }}
```

containers:

- name: nginx

```
image: "{{ .Values.image.repository }}:{{ .Values.image.tag }}"
```

15. Jenkins Docker Pipeline

Question

Write Jenkins pipeline for Docker build.

Answer

```
pipeline {  
  
    agent any  
  
    stages {  
  
        stage('Build Docker Image') {  
  
            steps {  
                sh 'docker build -t myapp .'  
            }  
        }  
  
        stage('Push Image') {
```

```
steps {  
  sh 'docker push myapp'  
}  
}  
}  
}
```

16. Real-Time Troubleshooting

Question

How do you troubleshoot failed Kubernetes deployment?

Answer

First, I check deployment status and pod events.

Then I verify container logs and image configuration.

I also check service connectivity and resource limits.

If required, I perform rollback.

17. Live Command Question

Question

What commands do you use for troubleshooting?

Answer

kubectl get pods

kubectl describe pod pod-name

kubectl logs pod-name

kubectl top pods

kubectl get events

18. Scenario Question

Question

How do you expose application externally in Kubernetes?

Answer

Services and Ingress are used to expose applications externally.

LoadBalancer service type is commonly used in cloud environments.

Ingress provides advanced routing and SSL support.

19. Scenario Question

Question

How do you secure Kubernetes secrets?

Answer

Secrets should not be hardcoded in YAML files.

Access permissions should be restricted using RBAC.

Secret management tools improve security further.

20. Production Best Practice

Question

What best practices do you follow for Kubernetes YAML files?

Answer

Proper labels and namespaces are used.

Resource limits and probes are configured.

Configurations are version controlled using Git.

YAML validation is performed before deployment.

Day 19 — EKS Complete Architecture + End-to-End Deployment Questions

Today focus on:

- AWS EKS architecture
- end-to-end deployment flow
- CI/CD integration
- real production workflow
- monitoring
- troubleshooting

These are very important for experienced DevOps interviews.

Speak slowly.

Pause after every sentence.

1. EKS Architecture

Question

Explain your EKS architecture.

Answer

Our application is deployed on AWS EKS.

We use a three-tier architecture containing frontend, backend, and database services.

The infrastructure is deployed inside a VPC with public and private subnets.

Load balancers are deployed in public subnets.

Worker nodes and applications run in private subnets.

Route53 is used for DNS management.

CloudWatch, Prometheus, and Grafana are used for monitoring.

2. End-to-End Deployment Flow

Question

Explain your end-to-end deployment process.

Answer

Developers push code into Git repositories.

Jenkins pipeline is triggered automatically.

The application build process starts.

Docker images are created and pushed to container registry.

Kubernetes manifests or Helm charts are updated.

ArgoCD detects Git changes and deploys applications to EKS clusters.

Monitoring tools verify deployment health after release.

3. Jenkins Pipeline Flow**Question**

Explain your Jenkins pipeline stages.

Answer

First, source code is pulled from Git.

Then the application build process starts.

Unit tests and vulnerability scans are executed.

Docker image is created and pushed to registry.

Deployment manifests are updated and synced through ArgoCD.

4. AWS ECR**Question**

What is ECR?

Answer

ECR stands for Elastic Container Registry.

It is a Docker image registry service provided by AWS.

Docker images are securely stored and managed in ECR.

5. Worker Nodes**Question**

What are worker nodes in EKS?

Answer

Worker nodes are EC2 instances that run Kubernetes workloads.

Pods are scheduled on worker nodes based on resource availability.

6. Control Plane**Question**

What is the EKS control plane?

Answer

The control plane manages Kubernetes cluster operations.

AWS manages the EKS control plane automatically.

It includes API Server, Scheduler, and Controller Manager.

7. Kubernetes Namespace

Question

How do you manage multiple environments in EKS?

Answer

Separate namespaces are created for development, staging, and production.

RBAC and resource quotas improve isolation and security.

8. Monitoring Architecture

Question

How do you monitor EKS clusters?

Answer

Prometheus collects metrics from Kubernetes clusters.

Grafana provides dashboards and visualization.

CloudWatch monitors AWS infrastructure and logs.

Alerts are configured for critical issues.

9. Logging Architecture

Question

How do you manage logs in Kubernetes?

Answer

Application and container logs are centralized using logging solutions.

Logs help identify application and infrastructure issues quickly.

10. Security

Question

How do you secure EKS clusters?

Answer

IAM roles and RBAC control access permissions.

Network policies restrict pod communication.

Secrets are used for sensitive information.

Container image scanning improves security.

11. Trivy

Question

How are you using Trivy?

Answer

Trivy is used for vulnerability scanning of Docker images.

It is integrated into Jenkins pipelines before deployment.

Images with critical vulnerabilities are blocked from deployment.

12. Ingress Controller

Question

Which ingress controller are you using?

Answer

We use NGINX ingress controller or AWS Load Balancer Controller.

It manages external application access and SSL routing.

13. HPA

Question

How are you handling scaling in Kubernetes?

Answer

Horizontal Pod Autoscaler automatically scales pods based on CPU or memory usage.

Cluster Autoscaler scales worker nodes when required.

14. Backup Strategy

Question

How do you manage Kubernetes backups?

Answer

Important configurations and databases are backed up regularly.

etcd backup is important for cluster recovery.

S3 is commonly used for backup storage.

15. Deployment Failure

Question

What will you do if deployment fails in production?

Answer

First, I check deployment events and pod logs.

Then I verify recent configuration and image changes.

If required, I rollback to the previous stable version.

Monitoring is used to validate application stability after recovery.

16. Production Troubleshooting

Question

How do you troubleshoot production issues in EKS?

Answer

First, I analyze monitoring alerts and logs.

Then I verify pod health, node status, and networking.

I also check recent deployment changes and resource utilization.

Root cause analysis is performed after issue resolution.

17. Real-Time Incident

Question

Tell me about a critical issue you handled in EKS.

Answer

One time, application pods were restarting continuously after deployment.

I checked pod logs and identified incorrect environment variable configuration.

I updated the configuration and restarted the deployment.

After validation, the application became stable.

18. Deployment Strategy

Question

Which deployment strategies are you using?

Answer

We use rolling updates for normal deployments.

Canary and blue-green deployments are used for critical applications.

These strategies reduce deployment risk and downtime.

19. Infrastructure Automation

Question

How are you automating infrastructure?

Answer

Terraform is used for infrastructure provisioning.

Infrastructure configurations are version controlled using Git.

Automation improves consistency and reduces manual errors.

20. DevOps Best Practice

Question

What DevOps best practices do you follow?

Answer

CI/CD automation improves deployment speed.

Infrastructure is managed as code.

Monitoring and security are integrated into workflows.

Rollback and backup strategies improve reliability.

21. Architecture Drawing Question**Question**

How will you explain your architecture in interview?

Answer

Users access the application through Route53 and Load Balancer.

Traffic reaches Kubernetes ingress controller.

Ingress routes traffic to frontend and backend services.

Applications run inside EKS worker nodes.

Databases run securely in private networks.

Monitoring and logging tools continuously monitor the environment.

22. Leadership Question**Question**

Have you handled deployments independently?

Answer

Yes, I have handled deployments, monitoring, and production troubleshooting independently.

I also coordinate with developers and QA teams during releases.

Day 20 — Complete Mock Interview Round (Technical + HR + Scenario)

Today practice like a real interview.

Rules:

- Speak slowly
- Pause after every sentence
- Give structured answers
- Stay calm

This round is designed for:

- 5 years DevOps Engineer
 - AWS + Kubernetes + CI/CD profile
-

1. Self Introduction**Question**

Please introduce yourself.

Answer

Hello, my name is Vipin Kumar.

I have around 5 years of experience as a DevOps Engineer.

I have hands-on experience with AWS, Kubernetes, Docker, Jenkins, Terraform, Git, ArgoCD, and Linux.

Currently, I am working on a three-tier application deployment project on AWS EKS.

My responsibilities include infrastructure provisioning, CI/CD automation, Kubernetes deployment, monitoring, troubleshooting, and production support.

I am looking for a new opportunity to improve my technical skills and career growth.

2. Current Project

Question

Explain your current project.

Answer

Currently, I am working on a three-tier application deployed on AWS EKS.

The architecture contains frontend, backend, and database services.

Developers push code to Git repositories.

Jenkins pipelines build Docker images and push them to ECR.

Kubernetes manifests and Helm charts are managed through Git repositories.

ArgoCD automatically deploys applications to EKS clusters.

Monitoring is handled using Prometheus, Grafana, and CloudWatch.

3. Kubernetes

Question

What happens when you create a Deployment in Kubernetes?

Answer

When a deployment is created, Kubernetes API Server receives the request.

Deployment controller creates a ReplicaSet.

ReplicaSet creates the required pods.

Scheduler assigns pods to worker nodes.

Kubelet starts containers on the assigned nodes.

4. Docker

Question

What is the difference between Docker image and container?

Answer

Docker image is a template containing application code and dependencies.

Docker container is a running instance of the image.

Multiple containers can be created from a single image.

5. Terraform

Question

How do you manage infrastructure using Terraform?

Answer

Terraform configuration files are stored in Git repositories.

We use terraform init, plan, and apply commands for infrastructure provisioning.

Remote backend is used for centralized state management.

Terraform modules improve reusability and consistency.

6. Jenkins**Question**

Explain your Jenkins pipeline.

Answer

Jenkins pipeline starts automatically after code is pushed to Git.

Source code is downloaded and application build process starts.

Docker image is created and scanned using Trivy.

Image is pushed to ECR.

Deployment manifests are updated and synced through ArgoCD.

7. ArgoCD**Question**

Why are you using ArgoCD?

Answer

ArgoCD supports GitOps-based deployment automation.

Git acts as the single source of truth.

ArgoCD continuously syncs Kubernetes clusters with Git repositories.

It improves deployment consistency and rollback management.

8. Troubleshooting**Question**

A pod is in CrashLoopBackOff state. How will you troubleshoot?

Answer

First, I check pod logs using kubectl logs command.

Then I verify environment variables and application configuration.

I also check database connectivity and resource limits.

After identifying the root cause, I fix the issue and restart deployment.

9. AWS Networking**Question**

What is the difference between Security Group and NACL?

Answer

Security Group works at the instance level and acts as a stateful firewall.

NACL works at the subnet level and acts as a stateless firewall.

10. Monitoring

Question

How do you monitor production applications?

Answer

Prometheus collects application and infrastructure metrics.

Grafana provides monitoring dashboards.

CloudWatch monitors AWS resources and alerts.

Logs and alerts help identify issues quickly.

11. Deployment Strategy

Question

Which deployment strategies have you used?

Answer

I have worked with rolling update, canary, and blue-green deployment strategies.

These strategies reduce deployment downtime and risk.

12. Security

Question

How do you secure Kubernetes clusters?

Answer

RBAC controls user permissions.

Secrets manage sensitive information.

Network policies restrict unnecessary communication.

Container image scanning improves security.

13. Linux

Question

How do you troubleshoot high CPU usage in Linux?

Answer

First, I check running processes using top command.

Then I identify high CPU-consuming applications.

I analyze logs and resource utilization.

After identifying the issue, I optimize or restart services if required.

14. Real-Time Scenario

Question

Production deployment failed. What will you do?

Answer

First, I check deployment logs and monitoring alerts.

Then I identify recent changes and root cause.

If the issue is critical, I perform rollback immediately.

After fixing the issue, I redeploy and monitor application stability.

15. High Traffic

Question

How do you handle high traffic in Kubernetes?

Answer

Horizontal Pod Autoscaler increases pod replicas automatically.

Cluster Autoscaler adds worker nodes if required.

Load balancers distribute traffic efficiently across pods.

16. Why Job Change

Question

Why are you looking for a change?

Answer

I have learned many technologies in my current organization.

Now I am looking for better growth opportunities and new technical challenges.

I want to work on large-scale cloud and DevOps projects.

17. Strength

Question

What are your strengths?

Answer

My strengths are troubleshooting, automation, and handling production issues calmly.

I also learn new technologies quickly and work well with teams.

18. Weakness

Question

What is your weakness?

Answer

Earlier, I was not very confident in interview communication.

Now I am improving my speaking skills through regular practice and mock interviews.

19. Leadership

Question

Have you handled responsibilities independently?

Answer

Yes, I have independently handled deployments, troubleshooting, and production support activities.

I also coordinate with development and QA teams during releases.

20. Final Question

Question

Do you have any questions for us?

Answer

Yes.

I would like to know more about the team structure and current projects.

I also want to understand the technologies and deployment processes used in this role.

1. Architecture Decision

Question

Why did you choose Kubernetes for your project?

Answer

Kubernetes provides scalability, high availability, and container orchestration.

It helps automate deployment, scaling, and self-healing.

Kubernetes is suitable for microservices and cloud-native applications.

It also improves deployment consistency across environments.

2. Cloud Decision

Question

Why are you using AWS instead of on-premises infrastructure?

Answer

AWS provides scalability, flexibility, and managed services.

Infrastructure provisioning becomes faster using cloud services.

AWS also improves availability, monitoring, and disaster recovery capabilities.

3. GitOps Decision

Question

Why did you implement GitOps?

Answer

GitOps improves deployment automation and consistency.

Git becomes the single source of truth for infrastructure and deployments.

Rollback and auditing become easier through Git history.

4. Production Downtime

Question

How do you reduce production downtime during deployment?

Answer

We use rolling updates, canary deployments, and load balancing.

Health checks ensure traffic reaches only healthy pods.

Rollback plans are prepared before deployment.

Monitoring is used continuously during release activities.

5. High Availability

Question

How do you ensure high availability in your architecture?

Answer

Applications run with multiple replicas across worker nodes.

Load balancers distribute traffic efficiently.

Multi-AZ deployment improves fault tolerance.

Monitoring and auto scaling improve reliability further.

6. Security Approach

Question

How do you handle security in DevOps environments?

Answer

IAM and RBAC control access permissions.

Secrets are managed securely.

Container image scanning is integrated into CI/CD pipelines.

Monitoring and auditing improve security visibility.

7. Production Incident

Question

Tell me about a critical production issue you handled.

Answer

One time, application pods started failing after deployment.

I checked logs and identified a configuration issue related to database connectivity.

I performed rollback immediately and restored services.

After fixing the configuration, deployment was completed successfully.

8. Team Coordination

Question

How do you coordinate with development teams during incidents?

Answer

I communicate issue impact and troubleshooting progress clearly.

We collaborate to identify root cause and deployment changes.

Regular updates help reduce confusion during incidents.

9. Infrastructure Scaling

Question

How do you handle infrastructure scaling?

Answer

Kubernetes HPA scales application pods automatically.

Cluster Autoscaler adds worker nodes when required.

AWS Auto Scaling improves infrastructure scalability further.

10. Cost Optimization

Question

How do you optimize AWS infrastructure cost?

Answer

We monitor resource utilization regularly.

Unused resources are cleaned up.

Auto scaling and right-sized instances improve cost optimization.

S3 lifecycle policies also reduce storage costs.

11. Monitoring Strategy

Question

What monitoring strategy do you follow?

Answer

We monitor infrastructure, applications, and Kubernetes clusters continuously.

CPU, memory, network, and response time metrics are tracked.

Alerts are configured for critical thresholds.

Logs and dashboards help identify issues quickly.

12. Disaster Recovery

Question

How do you handle disaster recovery?

Answer

Regular backups are maintained for important systems.

Multi-AZ and backup strategies improve recovery capability.

Disaster recovery procedures are tested periodically.

13. CI/CD Best Practice**Question**

What best practices do you follow in CI/CD pipelines?

Answer

Pipelines are automated and version controlled.

Security scanning and testing are integrated before deployment.

Rollback plans are prepared for production releases.

Access permissions are restricted properly.

14. Kubernetes Failure**Question**

What happens if a Kubernetes worker node fails?

Answer

Kubernetes automatically reschedules pods to healthy nodes.

ReplicaSets ensure the required number of pods remain running.

Monitoring alerts help identify node failures quickly.

15. DevOps Culture**Question**

What is DevOps culture according to you?

Answer

DevOps improves collaboration between development and operations teams.

It focuses on automation, faster delivery, and continuous improvement.

Monitoring, feedback, and reliability are also important parts of DevOps culture.

16. Leadership Question**Question**

Have you mentored junior engineers?

Answer

Yes, I have supported junior team members during deployments and troubleshooting.

I also help explain Kubernetes, CI/CD, and infrastructure concepts.

17. Handling Pressure

Question

How do you handle pressure during production incidents?

Answer

I stay calm and focus on identifying the root cause.

I prioritize critical tasks and communicate clearly with teams.

Structured troubleshooting helps reduce pressure during incidents.

18. Technical Decision**Question**

Why did you choose Helm in Kubernetes?

Answer

Helm simplifies Kubernetes deployments and configuration management.

It improves reusability and consistency across environments.

Managing multiple applications becomes easier using Helm charts.

19. Learning Ability**Question**

How do you keep yourself updated with new technologies?

Answer

I practice hands-on labs and follow technical documentation.

I also watch technical videos and explore real project use cases.

Continuous learning is very important in DevOps.

20. Final Manager Round Question**Question**

Why should we hire you?

Answer

I have strong hands-on experience in AWS, Kubernetes, CI/CD, and troubleshooting.

I have worked on real production environments and handled deployment issues.

I am also a quick learner and work well with teams.

I can contribute effectively to DevOps automation and infrastructure management.